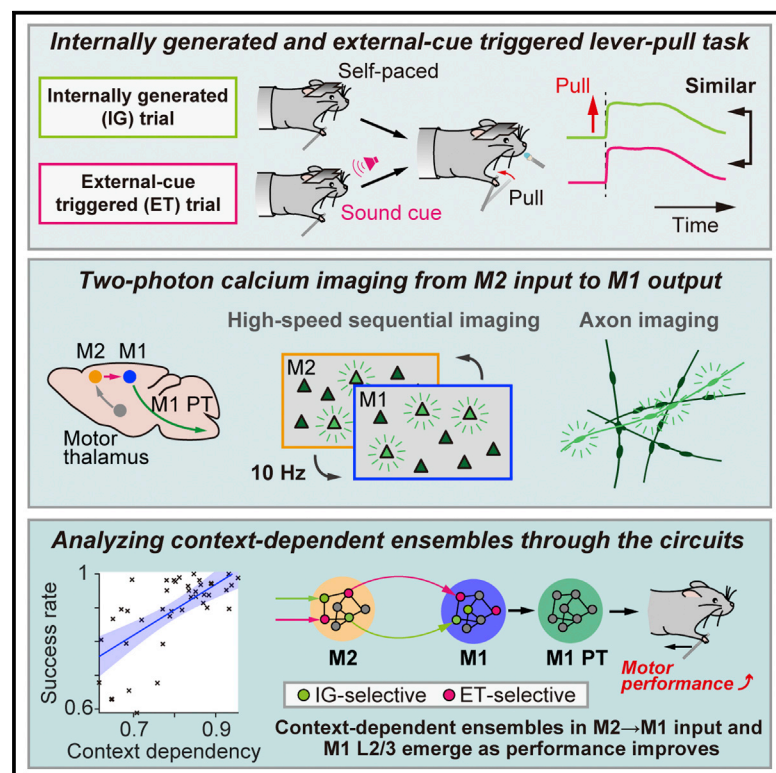


Transition of distinct context-dependent ensembles from secondary to primary motor cortex in skilled motor performance

Graphical abstract



Authors

Shin-Ichiro Terada, Kenta Kobayashi, Masanori Matsuzaki

Correspondence

mzakim@m.u-tokyo.ac.jp

In brief

Terada et al. conduct two-photon calcium imaging to reveal how different behavioral contexts are converted into the same goal-directed motor output in the motor cortex. They find that context-motor transformation occurs dynamically in M2 → M1 axons and M1 L2/3, and that it is associated with improvement in motor performance.

Highlights

- Head-fixed mice can perform similar IG and ET lever-pull movements within a session
- IG- and ET-dependent M2 L2/3 ensembles show large and consistent differences
- Context-dependent ensembles in M2 → M1 input and M1 L2/3 emerge as performance improves
- The stability of context-dependent M1 L2/3 ensembles is related to skilled movement



Article

Transition of distinct context-dependent ensembles from secondary to primary motor cortex in skilled motor performance

Shin-Ichiro Terada,¹ Kenta Kobayashi,² and Masanori Matsuzaki^{1,3,4,5,*}¹Department of Physiology, Graduate School of Medicine, The University of Tokyo, Tokyo 113-0033, Japan²Section of Viral Vector Development, National Institute for Physiological Sciences, Aichi 444-8585, Japan³International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo Institutes for Advanced Study, Tokyo 113-0033, Japan⁴Brain Functional Dynamics Collaboration Laboratory, RIKEN Center for Brain Science, Saitama 351-0198, Japan⁵Lead contact*Correspondence: mzakim@m.u-tokyo.ac.jp
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SUMMARY

When voluntary movements are executed under different contexts, different context-dependent signals are thought to weaken from secondary motor cortex (M2) to primary motor cortex (M1). However, it is unclear how the different contexts are processed from M2 to M1 to execute skilled movement. We conduct two-photon calcium imaging of M2 and M1 in mice performing internally generated and external-cue-triggered movements. Context dependency is consistently high in M2 L2/3 neurons and consistently low in M1 pyramidal tract neurons. By contrast, context dependency in M2 → M1 axons and M1 L2/3 neurons increases as task performance improves. In addition, the context dependency of M1 L2/3, but not M2 → M1 axons, is associated with fine-movement proficiency. The increase in context dependency correlates with stabilization of the context-dependent population activity and an increase in the neurons that strongly encode contextual and motor information. Thus, emergence of distinct context-dependent ensembles may be necessary for the context-to-motor transformation that facilitates skilled motor performance.

INTRODUCTION

The initiation of voluntary movement can be divided into internally generated (IG) and externally triggered (ET) types. These two behavioral contexts have been repeatedly contrasted over the decades because they are at the intersection of critical issues in neuroscience, such as sensorimotor transformation, motor planning, and intention (Haggard, 2008; Okano and Tanji, 1987; Passingham et al., 2010). In human and nonhuman primate neocortex, IG and ET contexts are represented in the supplementary motor area (SMA) and premotor area (PM) (Larsson et al., 1996; Okano and Tanji, 1987; Romo and Schultz, 1992). By contrast, activity in the primary motor cortex (M1) is similar between IG and ET types, and follows SMA and PM activity (Cunnington et al., 2002; Okano and Tanji, 1987; Riehle, 1991; Wildgruber et al., 1997). In the macaque, the SMA ensemble is involved in more context information and less muscle-related activity than the PM and M1 ensembles (Lara et al., 2018). In rodents, the secondary motor cortex (M2) (which is thought to correspond to the SMA and PM) and M1 strongly interconnect (Hooks et al., 2013; Ueta et al., 2014), and L2/3 ensembles in M2 and M1 show high trial-to-trial correlation in their activity during motor execution (Terada et al., 2018; Veuthy et al., 2020). M2 population activity precedes M1 population activity, and

M2 activity is required for the M1 activity needed to execute the appropriate movement (Veuthy et al., 2020). The activity flow from M2 to M1 emerges during learning of sensorimotor tasks (Esmaili et al., 2021; Makino et al., 2017). Thus, top-down control dominates in the motor cortex, with M2 and M1 tightly interacting. However, it is unknown which types of information are transmitted from M2 to M1 when the same movement is triggered by different behavioral contexts, such as IG and ET types.

In M1, L2/3 ensembles dynamically change according to the stage of motor learning (Huber et al., 2012; Masamizu et al., 2014; Peters et al., 2014). M1 L2/3 activity flexibly reorganizes to execute appropriate locomotion when animals are presented with unexpected visual perturbation (Heindorf et al., 2018). Through the learning of locomotion in different environmental contexts (regularly and irregularly spaced ladder rungs), context-dependent ensembles in M1 L2/3 emerge, and the silencing of projections from M2 to M1 decreases the context-dependent coding of movement by M1 L2/3 neurons (Omlor et al., 2019). In contrast to M1 L2/3 neurons, M1 layer 5 (L5) neurons directly represent the executed movement (Heindorf et al., 2018; Masamizu et al., 2014). M1 L2/3 neurons have the ability to directly drive motor output neurons in L5 (Ueta et al., 2014; Weiler et al., 2008). Thus, we hypothesized that M1 L2/3 neurons



receive some context-dependent signals that are possessed by M2 ensembles and transform them to drive L5 neurons to effectively output motor commands. However, it is unknown whether and how the context-dependencies of the ensembles in M2, M2 → M1 projections, and M1 are related to the skilled motor performance. One possibility is that M2 neurons continue to send strong context signals to M1 and, on the basis of these signals, only M1 L2/3 neurons reorganize to achieve the skilled motor performance. In the visual and somatosensory cortex, the signal transmitted from the primary sensory area to the higher-order sensory area is transformed to the type of signal that is present in the target area (Chen et al., 2013a; Glickfeld et al., 2013). If there is an analogy with this, the context signals of M2 → M1 projections may be selected in conjunction with a change in those of M1 L2/3.

To address these issues, we trained head-fixed mice to perform an IG-ET task that consisted of IG and ET lever-pull trials in the same sessions (Masamizu et al., 2014; Tanaka et al., 2018; Terada et al., 2018). To compare L2/3 neuronal activities between M2 and M1 and reveal their interactions at population and single-neuron levels, it is crucial to measure the activity of multiple neurons in the same trials because trial-to-trial variability in behavior affects neuronal activity (Musall et al., 2019; Stringer et al., 2019). Therefore, we used super-wide-field two-photon microscopy, which can sequentially image forelimb M2 and M1 (we refer to them simply as M2 and M1, respectively) that are 2 mm apart (Terada et al., 2018). To reveal the signal transmitted from M2 to M1, we imaged the activities of axons from M2 to M1 layer 1 (L1; M2 → M1 axons). Although the difference in the behavioral contexts was subtle, we found that M2 L2/3 constantly coded strong context signals regardless of how well the animal performed the task. By contrast, context-dependent ensembles emerged in the M2 → M1 axons as the success rate and reaction time in ET trials improved, but the information on lever-pull movement encoded in individual M2 → M1 axons did not increase. In M1 L2/3 context-dependent ensembles emerged, and the information on lever-pull movement encoded in individual neurons increased as the performance parameters and fine-movement proficiency improved.

RESULTS

M2 and M1 are required for IG and ET lever pulls

In IG trials, head-fixed mice needed to wait at least 2.6 s before using their right forelimb to pull a lever and hold it for 400 ms to obtain a drop of water (Figure 1A). No sound cue was delivered. In ET trials (Terada et al., 2018), the mice had to pull the lever within 500 ms after the onset of a 100-ms sound cue (Figure 1A). The sound cue was delivered 2.5–3.5 s after the lever was returned in the preceding lever pull. In the IG-ET task, the blocks of the IG and ET trials (IG block and ET block, respectively) were switched every 3 min during each session (Figure 1B). Mice were first trained to perform only the ET trials for 10–14 training sessions, and were then trained on the IG-ET task for at least ten training sessions. After these training sessions, we conducted two-photon calcium imaging during the IG-ET task.

To confirm that M2 and M1 were necessary for the IG-ET task even after the training sessions, we injected the GABA_A receptor

agonist muscimol into M2 or M1 (Figures S1A and S1B). Inactivation of either area substantially reduced the lever-pull performance in a similar manner: the number of rewards and the number of pull initiations in IG trials, and the number of rewards, reaction time, and success rate in ET trials (Figures 1C and S1C). M1 inactivation also caused a monoparesis-like symptom in the right forelimb, and the forepaw was not placed on the lever in nearly 80% of ET trials, whereas no such symptom was observed during M2 inactivation (Figure S1D). This is consistent with recent reports (Currie et al., 2022; Morandell and Huber, 2017).

However, since the effect of the muscimol inactivation was maintained during the task, it was unclear whether M2 and M1 were required for the lever-pull movement after the lever-pull initiation. To address this issue, stGtACR1, a photoactivatable chloride channel (Mahn et al., 2018), was expressed in M2 and M1 excitatory neurons, after which an orange laser was illuminated on M2 or M1 for 1 s immediately after the lever pull started (Figures 1D and S1E–S1G). Light illumination on M1, but not on M2, induced a significantly large displacement of the lever trajectory in both IG and ET trials, although the illumination condition was the same between M2 and M1 (Figures 1E and 1F). Even when the laser intensity was doubled for the M2 illumination, the lever trajectory did not apparently deteriorate (Figure S1H). The latency of the displacement of the lever trajectory in the M1 illumination was 25.5 ± 0.55 ms ($n = 8$ sessions from four mice). This short latency suggests direct control of the forelimb movement from M1 to the spinal cord (Miri et al., 2017). Thus, after the lever-pull onset, M1 was more important for motor execution than M2.

Two-photon calcium imaging of M2 and M1 neurons during the IG-ET task

To compare the neuronal activity in L2/3 between M2 and M1 during the IG-ET task, we used super-wide-field two-photon microscopy (Terada et al., 2018). This microscopy involves a pair of micromirrors positioned in the optical pathway below the objective lens and above the cranial window. By rotating a small device attached to the mirror pair back and forth, it is possible to switch the field of view between distant areas (Figures S2A and S2B). We improved the rotation and supporting mechanisms of the device so that M2 and M1 fields could be imaged alternately at a frequency of 10 Hz with minimal vibration from the rotation (Figures S2A–S2H). We sequentially imaged RCaMP1.07-expressing L2/3 fields in M2 and M1 for approximately 24 min in each IG-ET task session, and detected 60–262 active neurons in each field (Figures 2A and 2B).

Subsets of neurons apparently exhibited the activity that was selective to IG trials or ET trials (Figure 2C). First, according to the activity averaged over the lever-pull period (0.4 s before to 2 s after successful lever-pull onset), we classified the neurons as either task-related (the mean proportions were 0.82 [95% confidence interval [CI] 0.80–0.83] of the total in M2; 0.88 [CI 0.85–0.90] in M1; $n = 40$ sessions from six mice; see STAR Methods) or task-unrelated. Next, according to the differences in their time-averaged activity between IG and ET trials, task-related neurons were classified into three types: neurons that showed IG-type-dependent activity (“IG neurons;” mean proportions

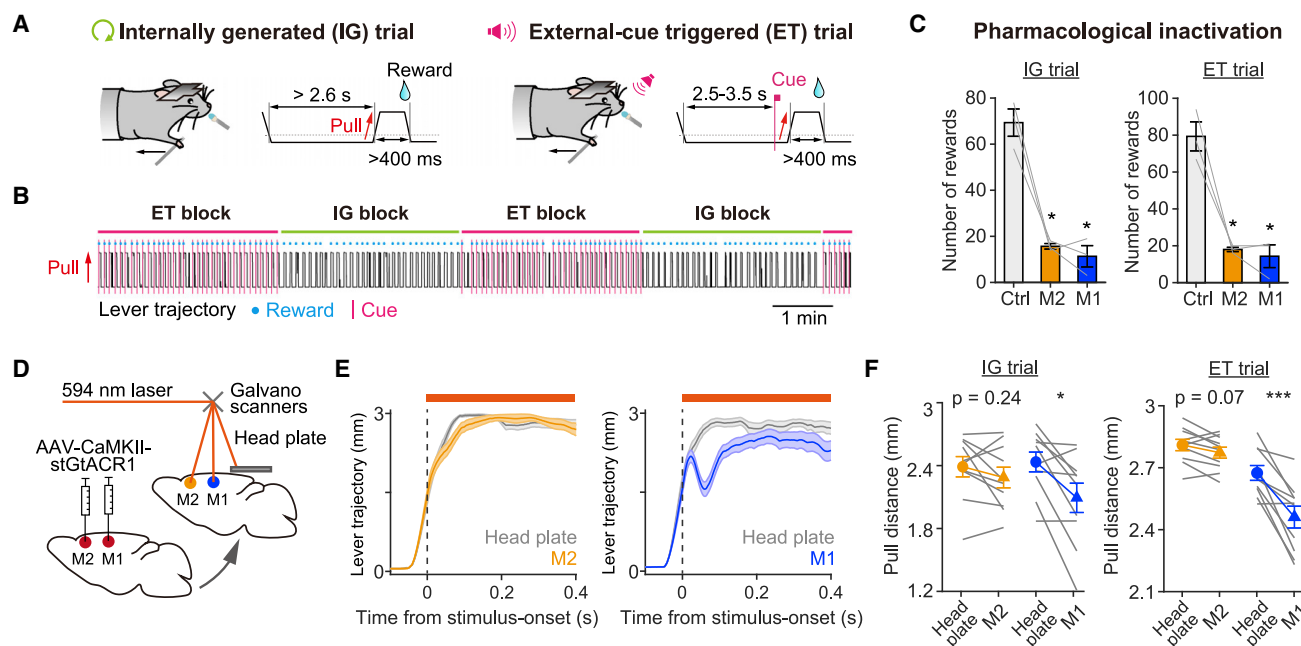


Figure 1. The IG and ET lever-pull task that required M2 and M1

(A) Schematic diagram of the IG-ET task.

(B) Representative performance in the IG-ET task, showing lever trajectories (black lines), reward timings (light blue dots), and sound cues (vertical magenta lines).

(C) Effect of muscimol injection into M2 or M1 on the number of rewarded trials. Saline was injected as a control. Error bars, SEM. * $p < 0.05$, paired t test, $n = 3$ mice.

(D) Schematic illustration of the optogenetic experiment.

(E) Trial-averaged lever trajectory during the light illumination. Laser irradiation (top dark-orange bar) was initiated at the lever-pull onset in ET trials. In one session, either M2 only (orange) or M1 only (blue) was the stimulus target, and the head plate (gray) was the control target. Two representative sessions from the same mouse are shown. Shading represents the SEM. Left, $n = 28$ and 40 trials for M2 and head plate, respectively. Right, $n = 35$ and 41 trials for M1 and head plate, respectively.

(F) Left: effect of the light illumination on the pull distance in IG trials. The pull distance was the averaged lever position from 0 to 0.4 s from light illumination onset. * $p < 0.05$, paired t test, $n = 10$ sessions from five mice for M2 and $n = 10$ sessions from five mice for M1. Right: same plot as in the left panel but for ET trials. Error bars, SEM. *** $p < 0.001$, paired t test, $n = 10$ sessions from five mice for M2 and $n = 10$ sessions from five mice for M1.

See also Figure S1.

0.21 [CI 0.19–0.24] in M2; 0.18 [CI 0.15–0.22] in M1; $n = 40$ sessions from six mice), neurons that showed ET-type-dependent activity (“ET neurons,” mean proportion 0.15 [CI 0.13–0.17] in M2; 0.093 [CI 0.072–0.12] in M1; $n = 40$ sessions from six mice), and other neurons (“common neurons,” defined as neurons that showed common activity in both trial types; mean proportions 0.64 [CI 0.61–0.67] in M2; 0.73 [CI 0.68–0.77] in M1; $n = 40$ sessions from six mice; see STAR Methods) (Figures 2D and 2E). The activity of IG and ET neurons occurred mainly around the lever-pull onset, with the IG-type-selective activity starting earlier than the ET-type-selective activity (Figures 2D–2F). Those IG neurons with trial-averaged peak activity before the lever-pull onset showed a small ramping in activity toward lever-pull onset, and the amplitude of the pre-movement activity correlated with the waiting time in IG trials (Figures S3A and S3B). In most of the ET neurons, the response to the cue tone in non-lever-pull trials was much smaller than that in lever-pull trials (Figures 2G and 2H). These results suggest that the activity of IG and ET neurons contributed to the context-dependent lever-pull preparation, initiation, and movement. The proportions of IG and ET neurons were higher in M2 than in M1, although the proportions of IG neu-

rons were not significantly different (IG, $p = 0.16$; ET, $p < 10^{-4}$; paired t test, $n = 40$ sessions from six mice). There was no apparent deviation in the distribution of IG or ET neurons along the anterior-posterior and lateral-medial axes. However, the distance between neurons of the same type was less than that between different types of neurons in M2 and M1 (Figures S3C and S3D). This suggests that neurons of the same types were weakly clustered. In common neurons, the difference in the peak timing of the activity between IG and ET trials was much smaller than that between different common neurons (Figure 2I). Thus, the task-related activity of common neurons generally showed similar amplitude and peak timing in IG and ET trials.

Trial-type classification accuracy in different neuronal classes

Classification of IG, ET, and common neurons is useful for capturing the basic properties of the trial-type-dependent activity of individual neurons. However, as individual neurons are only contributors to the neuronal network, it is important to reveal the time-evolving population activity to understand the neural control of movement (Gallego et al., 2017; Golub et al., 2018; Shenoy

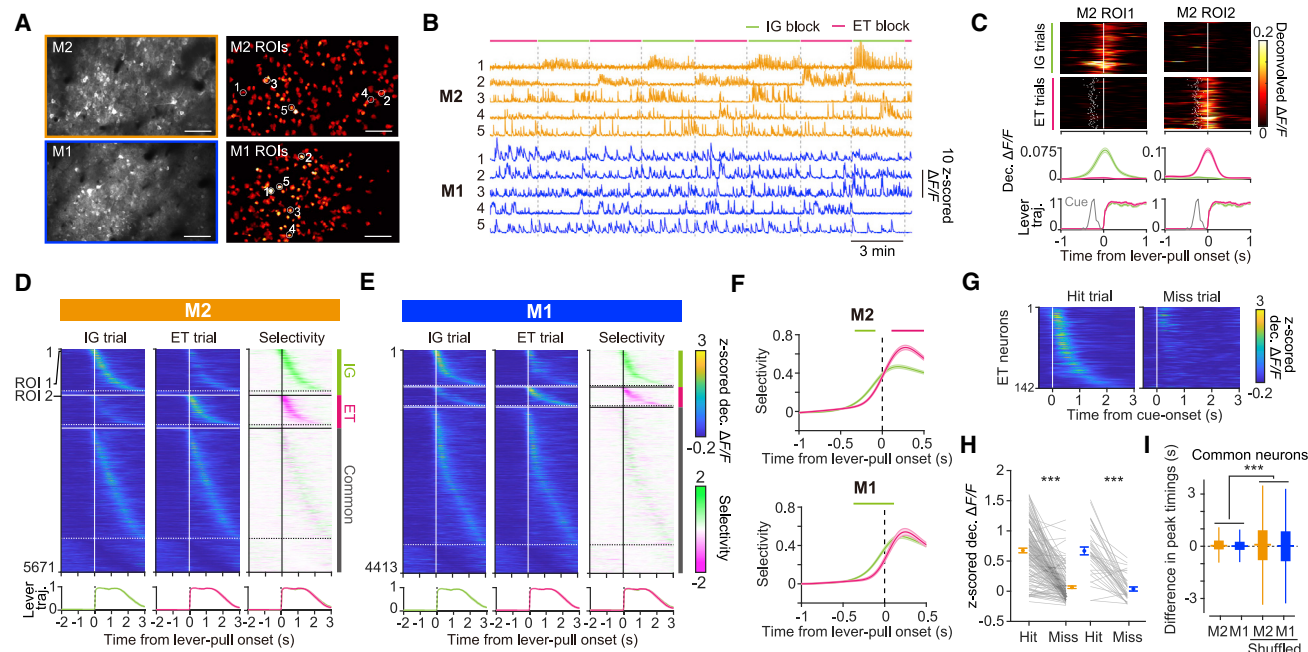


Figure 2. Subsecond sequential two-photon calcium imaging of M2 and M1 during the IG-ET task

(A) Left: frame-averaged two-photon images of L2/3 neuronal somata expressing R-CaMP1.07 in M2 and M1. Right: spatial components extracted by the constrained non-negative matrix factorization algorithm. Scale bar, 100 μ m.

(B) Representative $\Delta F/F$ traces of M2 and M1 L2/3 neurons during the IG-ET task. The numbers to the left of each trace correspond to the regions of interest (ROIs) shown in (A).

(C) Representative IG (left) and ET (right) neurons in M2 (ROIs 1 and 2 in B, respectively). From top to bottom, pseudo-color plots of the deconvolved (dec.) $\Delta F/F$ in successful IG trials, successful ET trials, trial-averaged deconvolved $\Delta F/F$ (green for IG trials, magenta for ET trials), and trial-averaged lever trajectories (traj.). White dots indicate the cue onset in the ET trials. Gray lines in the bottom panels show distributions of the cue onset.

(D and E) Time courses of trial-averaged activity and trial-type selectivity of task-related neurons in M2 L2/3 (D) and M1 L2/3 (E). Data are pooled from 40 sessions in six mice. Left: activity in IG trials; middle: activity in ET trials; right: trial-type selectivity. Bottom panels indicate the trial-averaged lever trajectories. The thick vertical lines on the right indicate the significance of the selectivity; green indicates IG neurons, magenta ET neurons, and gray common neurons. The data are sorted by the peak timing (see STAR Methods) of the activity within each cell type, and each row corresponds to the same neuron. White horizontal lines indicate the border between different types of neurons. Within each type, the neurons that increased activity during the lever-pull period are above the horizontal dotted line, and the neurons that showed decreased activity are below it. ROI 1 and ROI 2 correspond to the example neurons in (C).

(F) Time course of the averaged trial-type selectivity of IG (green) and ET (magenta) neurons in M2 and M1 (see STAR Methods). The shading represents the SEM. The green and magenta bars at the top are the time points where selectivity of IG or ET neurons was significantly higher than that of ET or IG neurons, respectively ($p < 0.05$, unpaired t test with Bonferroni correction). $n = 1,172$ neurons in 40 sessions from six mice for M2 IG neurons, and 837 neurons in 40 sessions from six mice for M2 ET neurons. $n = 727$ in 40 sessions from six mice for M1 IG neurons in M1, and 400 neurons in 40 sessions from six mice for M1 ET neurons.

(G) Trial-averaged activity of M2 ET neurons aligned to the cue onset of hit (left) and miss (right) ET trials. A miss trial was defined as a trial in which the lever did not exceed the threshold within 500 ms of the cue onset. Sessions with at least five miss ET trials were included in the analysis ($n = 9$ sessions from four mice).

(H) Comparison of the averaged Z-scored deconvolved $\Delta F/F$ from 0 to 1 s after the cue onset for hit and miss trials. *** $p < 0.001$, paired t test. Orange, the mean $\pm$ SEM of M2 ET neurons ($n = 142$ in nine sessions from four mice); blue, the mean $\pm$ SEM of M1 ET neurons ($n = 38$ in nine sessions from four mice).

(I) Box-and-whisker plots of the difference in peak timings between IG and ET trials for common neurons. Outliers are not shown. The values are compared with those calculated from peak timings shuffled between common neurons in each area. *** $p < 0.001$, Kolmogorov-Smirnov test, $n = 3,662$ in 40 sessions from six mice for M2, and 3,286 neurons in 40 sessions from six mice for M1.

See also Figures S2 and S3.

et al., 2013). Therefore, we estimated the time course of the representation of trial-type information by the neuronal population in M2 and M1. We trained a support vector machine to discriminate the type (IG or ET) of each trial in the training trials according to the activity patterns of 60 neurons and used this trained decoder to calculate the trial-type classification accuracy (TCA) in the test trials (see STAR Methods; Figures 3A and 3B). The TCA of M2 neurons started to increase to ~ 0.5 s before the lever-pull onset, peaked at >0.8 around the pull onset, and then gradually decreased (Figure 3C). M2 showed significantly higher TCA than M1 during the continuous period from ~ 0.5 s before to

~ 2.5 s after the onset of the lever pull (Figure 3C). We refer to the TCAs calculated from M2 and M1 population activity averaged over the pull-onset window (from 0.4 s before to 0.4 s after pull onset) as TCA_{M2} and TCA_{M1} , respectively. TCA_{M2} was higher than TCA_{M1} in most sessions ($p < 10^{-7}$, paired t test, $n = 40$ sessions from six mice; Figure 3D), and this difference did not depend on the number of neurons used for decoding (Figure S4O).

Next, we investigated the TCA upstream of M2 L2/3 neurons, between M2 and M1, and downstream of M1 L2/3 neurons. Parts of the motor thalamus (MTh), such as the ventral medial nucleus

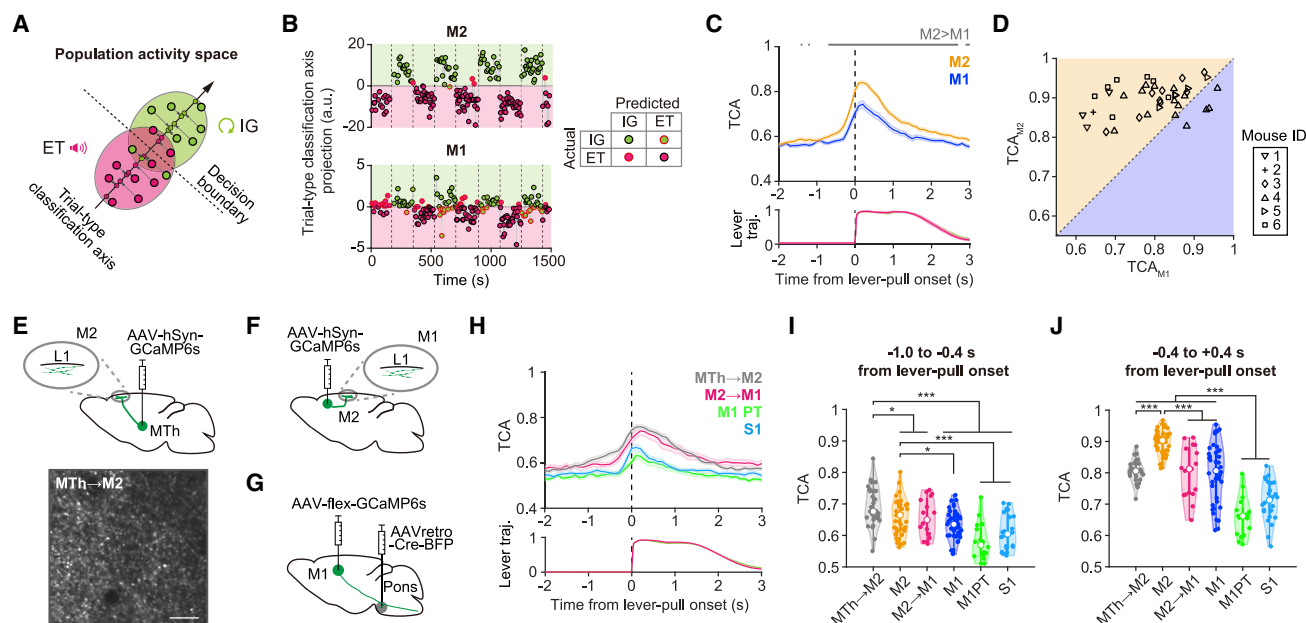


Figure 3. TCAs of different neuronal classes

(A) Schematic explanations of trial-type classification analysis. Each dot indicates the population activity of each trial in the hypothetical two-dimensional space. Green dots, IG trials; magenta dots, ET trials.

(B) M2 and M1 trial-type classification axis projection of all successful lever-pull trials in a representative session. x axis indicates the session time. Green dots, IG trials; magenta dots, ET trials. Dots with black and red edges are correctly and incorrectly labeled trials, respectively. Population activity averaged over the pull-onset window (from 0.4 s before to 0.4 s after the pull onset) was used.

(C) Time course of the TCAs of M2 (orange) and M1 (blue) neurons. The gray bars at the top are the time points at which M2 was significantly higher than M1 ($p < 0.05$, paired t test with Bonferroni correction; $n = 40$ sessions from six mice). Shading represents the SEM. The bottom row is the averaged lever trajectory.

(D) Scatterplot of TCA_{M2} and TCA_{M1} . Each dot indicates a separate session. Different symbols indicate different mice.

(E) Bottom: representative time-averaged two-photon images of MTh $\rightarrow$ M2 axons. Scale bar, 20 μ m.

(E–G) Schematics of the injection strategy for imaging MTh $\rightarrow$ M2 axons (E, top), M2 $\rightarrow$ M1 axons (F), and M1 PT neurons (G).

(H) Time course of the TCAs of MTh $\rightarrow$ M2 axons (gray, $n = 31$ sessions from six mice), M2 $\rightarrow$ M1 axons (magenta, $n = 17$ sessions from four mice), M1 PT neurons (green, $n = 17$ sessions from seven mice), and S1 L2/3 neurons (light blue, $n = 24$ sessions from five mice). Shading represents the SEM.

(I and J) Violin plots of TCA for each imaged structure calculated from the activity averaged over the time window from -1.0 to -0.4 s from lever-pull onset (I) and the pull-onset window (J). Median values are shown as open circles within each violin. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Wilcoxon rank-sum test.

See also Figures S4–S6.

(VM) and ventral anterior nucleus (VA), mainly receive outputs downstream of the dorsal striatum, which is important for the initiation of voluntary movements (Aoki et al., 2019; Catanese and Jaeger, 2021; Klaus et al., 2019). Subsets of these thalamic neurons send their axons to L1 in M2 (Kuramoto et al., 2015). Therefore, we conducted two-photon calcium imaging of MTh $\rightarrow$ M2 axons (Figures 3E and S4A). We also conducted two-photon calcium imaging of M2 $\rightarrow$ M1 axons that originated from L2/3 and L5 neurons in M2 (Tanaka et al., 2018; Terada et al., 2018; Ueta et al., 2014) (Figures 3F, S4B, and S4C). We also imaged the apical dendrites of PT neurons (Figures 3G and S3D–S3J) and forelimb S1 L2/3 neurons. For imaging of the axons and dendrites, we used GCaMP6s (see STAR Methods). As axon and dendrite imaging could include multiple boutons or dendrites originating from the same neurons (Petreanu et al., 2012; Ranganathan et al., 2018; Tanaka et al., 2018), the regions of interest (ROIs) with high activity correlation of >0.7 were considered to be from the same origin, and the ROI with the highest signal-to-noise ratio was used as a representative for the analysis.

We calculated the TCA from task-related axonal boutons, dendrites, and neuronal somata extracted from each type of neuron (Figures 3H and S4K–S4N). The TCA of the MTh $\rightarrow$ M2 axons rose slightly earlier than the others, and TCA calculated from the activity averaged over the time window from 1 s to 0.4 s before the lever-pull onset was highest in the MTh $\rightarrow$ M2 axons (Figures 3H and 3I). TCA was highest around the pull onset in each type of neuron, and TCA calculated from the activity averaged over the pull-onset window was highest in M2 L2/3 (Figures 3H and 3J). In the M2 $\rightarrow$ M1 axons and M1 L2/3 neurons, the TCA around the pull onset was lower than that in M2 L2/3 but higher than that of M1 PT neurons. In S1 L2/3 neurons, the TCA was as low as that of PT neurons (Figure 3J). These differences did not depend on the number of neurons used for decoding (Figure S4O). Even when small calcium responses were removed from the TCA calculation in M2 and M1 L2/3 neurons, the reduction in TCA was slight (Figures S4P and S4Q). Thus, it is unlikely that the TCAs of the imaged axons and dendrites were much more underestimated than the TCAs of M2 and M1 L2/3 neurons (see also STAR Methods). These results suggest that

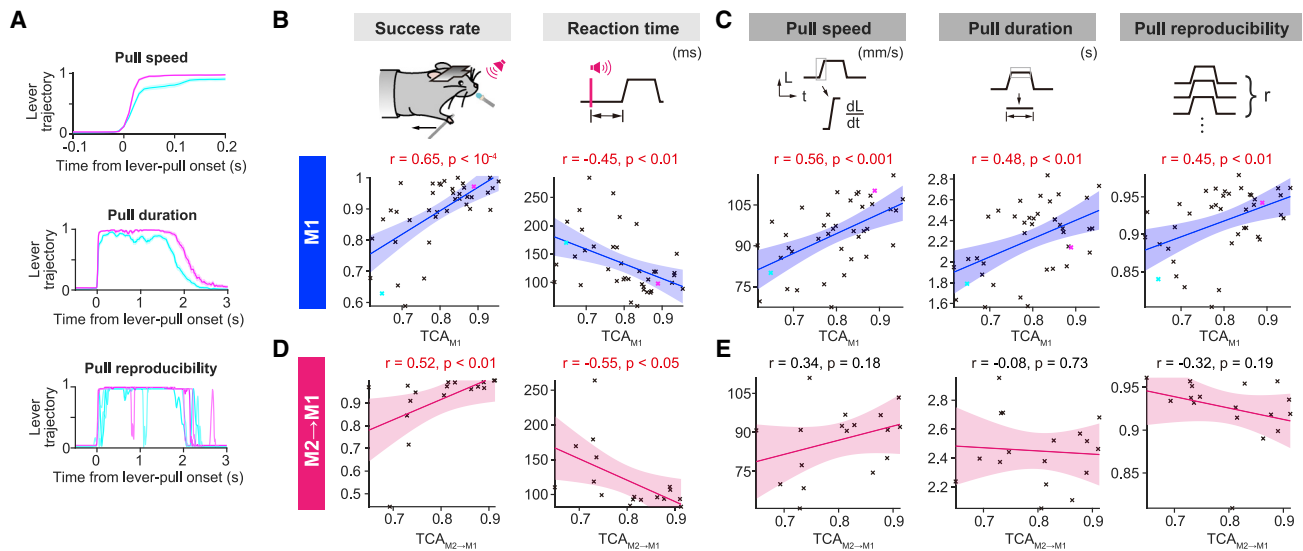


Figure 4. Relationships between motor performance and TCA_{M1} and $TCA_{M2 \rightarrow M1}$

(A) Representative lever-pull kinematics in a low (light blue) TCA_{M1} session and a high (magenta) TCA_{M1} session. For the pull speed and pull duration, the average lever trajectories of ET trials are plotted. For the pull reproducibility, five randomly selected ET trials are overlaid.

(B) Scatterplots of the success rate and the reaction time in ET trials against TCA_{M1} . Light-blue and magenta points correspond to the sessions shown in (A). r , Pearson's correlation coefficient, $n = 40$ sessions from six mice. Colored lines indicate the fit of linear models and shadows indicate 95% confidence intervals.

(C) Scatterplots of the three lever-pull kinematics against TCA_{M1} . $n = 40$ sessions from six mice. The kinematics were averaged across both IG and ET trials in each session. Colored lines indicate the fit of linear models and shadows indicate 95% confidence intervals.

(D and E) Same plots as in (B) and (C) but for $TCA_{M2 \rightarrow M1}$. $n = 17$ sessions from four mice. Colored lines indicate the fit of linear models, and shadows indicate 95% confidence intervals.

See also Figure S7.

the trial-type information gradually decreased from M2 L2/3 neurons to M1 output neurons.

As TCA_{M1} increases, task performance improves, and the speed and reproducibility of lever movement increases

In contrast to the constantly high TCA_{M2} (0.8–1), TCA_{M1} and TCAs calculated from $M2 \rightarrow M1$ axons averaged over the pull-onset window ($TCA_{M2 \rightarrow M1}$) showed high session-to-session and animal-to-animal variability (from 0.6 to 1; Figures 3J and S4R), and TCA_{M2} did not correlate with TCA_{M1} ($p = 0.12$, Pearson's correlation coefficient; $n = 40$ sessions from six mice; Figure 3D). What was related to the high variability in TCA_{M1} and $TCA_{M2 \rightarrow M1}$?

TCA that was calculated with a decoder using only one forelimb-related movement variable extracted from two videos was less than 0.6 (Figures S5A–S5D). By contrast, TCA calculated from the jaw movement averaged over the pull-onset window (TCA_{jaw}) was approximately 0.75 and correlated strongly with TCA_{M1} and $TCA_{M2 \rightarrow M1}$, but not with TCA_{M2} (Figures S5D and S5E). Jaw movement before the lever-pull onset (pre-pull jaw movement) was apparently more frequent in ET trials than in IG trials (Figure S5C). However, the proportion of pre-pull jaw-movement-related, pre-pull licking-related, and pre-pull nose-movement-related neurons (pre-pull orofacial-movement-related neurons; see STAR Methods for details) was approximately 0.07, and their overlap with ET neurons was small (Figure S5G). When we calculated TCA_{M1} and $TCA_{M2 \rightarrow M1}$ with the activity of pre-pull orofacial-

movement-related neurons shuffled among the test trials, TCA_{M1} or $TCA_{M2 \rightarrow M1}$ did not decrease (Figure S5H). By contrast, shuffling the activities of IG and ET neurons reduced TCA_{M1} and $TCA_{M2 \rightarrow M1}$ to approximately 0.6 and 0.65, respectively (Figure S5H). Thus, pre-pull orofacial-movement-related neurons were not directly related to TCA_{M1} or $TCA_{M2 \rightarrow M1}$. In addition, although inactivation of $tjM1$ (an area that is strongly related to jaw movement) (Mayrhofer et al., 2019) inhibited the pre-pull jaw movement, and pre-pull licking and TCA calculated from each orofacial movement decreased to less than 0.6, the lever-pull movement or jaw and licking movement for water consumption was not apparently inhibited (Figures S6A–S6G). Thus, the pre-pull jaw movement was not directly related to the lever-pull movement. The horizontal location of the imaging fields in M1 was not also related to TCA_{M1} (Figure S5I).

We next examined whether the variability of TCA_{M1} and $TCA_{M2 \rightarrow M1}$ over sessions might be related to the variability of the task performance, including fine-movement (lever-pull kinematic) proficiency (Figure 4A). TCA_{M1} showed strong positive correlation with success rate and negative correlation with reaction time in ET trials (Figure 4B). In addition, as TCA_{M1} increased, the fine-movement proficiency in both IG and ET trials combined increased: the pull speed, pull duration, and pull reproducibility (Tanaka et al., 2018; Terada et al., 2018) increased (Figures 4A and 4C), regardless of whether the trial was of the IG or ET type (Figure S7). Thus, as the trial-type-dependent activities showed better segregation, the task performance improved and stabilized in the same way as in IG and ET trials. $TCA_{M2 \rightarrow M1}$ correlated with

success rate and reaction time in ET trials, but not the fine-movement proficiency (Figures 4D and 4E). This suggests that the signals conveyed by segregated trial-type-dependent ensembles of the M2 → M1 axons played a role in the induction of successful pulling but not in fine lever-pull proficiency.

As TCA_{M1} increases, the trial-to-trial variability of M1 L2/3 population activity within each trial type decreases

There are two neuronal mechanisms to ensure good separation of IG-dependent and ET-dependent population activities in the population activity space in which each axis represents the activity of a single neuron (Golub et al., 2018): an increase in the distance between trial-averaged IG-dependent population activity and trial-averaged ET-dependent population activity, and a decrease in the trial-to-trial variability of the population activity within each type of IG and ET trial (Figure 5A). To test these possible mechanisms, we used the activity of individual neurons averaged over the pull-onset window to calculate the cosine distance in the trial-averaged vector of the population activity between the IG and ET trials as D_{IG-ET} , and the variance of the cosine distance between the trial-averaged vector and each trial vector of the population activity in IG or ET trials as V_{IG} or V_{ET} , respectively (Figure 5A).

In the M1 population activity, D_{IG-ET} became longer as TCA_{M1} increased, and both V_{IG} and V_{ET} became smaller than the corresponding values of M2 as TCA_{M1} increased (Figures 5B and 5C). In M2 → M1 axons, D_{IG-ET} also became longer as $TCA_{M2 \rightarrow M1}$ increased (Figure 5D), but V_{IG} and V_{ET} did not change (Figure 5D). In the M2 population activity, neither D_{IG-ET} , V_{IG} , nor V_{ET} depended on TCA_{M1} (Figure 5E). Thus, the separation of IG-dependent and ET-dependent population activities in the population activity space differed between M1 L2/3, M2 → M1 axons, and M2 L2/3. In addition, in M1, V_{IG} and V_{ET} , but not D_{IG-ET} , significantly correlated with success rate and reaction time in ET trials, and with the fine-movement proficiency (Figure 5F). By contrast, in M2 → M1 axons, only D_{IG-ET} strongly correlated with success rate and reaction time in ET trials (Figure 5F). These results suggest that longer D_{IG-ET} in M2 → M1 axons increased task performance in ET trials, while smaller V_{IG} and V_{ET} in M1 L2/3 increased task performance in ET trials and stabilized the fine lever-pull movement (Figure 5G).

Lever movement is strongly encoded by IG and ET neurons in M1 L2/3 but not in M2 → M1 axons

We then returned to the properties of individual neurons and asked the question “how is the increase in TCA_{M1} related to changes in the trial-type and motor information encoded in IG, ET, and common neurons in M1 L2/3?” First, we constructed an encoding model with the lever-movement variables (pull, hold, return, and speed), licking, jaw movement, and trial type (behavior model; Figures 6A and S8A). The model’s prediction accuracy (R^2) was much higher for the activity of task-related neurons than for task-unrelated neurons (Figures 6B, S8B, and S8C). Next, we estimated the unique contribution as the difference in R^2 between the behavior model and the model without target variable(s) (ΔR^2 ; Musall et al., 2019). The unique contributions of lever movement and trial-type variables to the activity of IG, ET, and common neurons in M1 L2/3 were higher in high TCA_{M1} (>0.8) sessions than in low TCA_{M1} (<0.8) sessions (Fig-

ure 6C), while the unique contribution of lever movement was higher in IG and ET neurons than in common neurons. The proportions of IG and ET neurons increased 1.6-fold and 3.7-fold, respectively (Figure 6D). The unique contribution of the trial type correlated with the lever movements in individual neurons, particularly IG and ET neurons (Figure S8D). Thus, the increase in IG and ET neurons would increase the association between the fine-movement proficiency and M1 population activity. By contrast, in the M2 → M1 axons, the unique contribution of lever movement to IG, ET, or common neurons was not significantly different between high and low $TCA_{M2 \rightarrow M1}$ sessions (Figure 6E). This single-neuron property might explain why $TCA_{M2 \rightarrow M1}$ did not correlate with the increase in proficiency of lever-pull movement, even though the proportions of IG and ET neurons increased as $TCA_{M2 \rightarrow M1}$ increased (Figure 6F). In M2 L2/3, the unique contribution of lever movement to IG, ET, or common neurons, or the proportions of either type of neurons, were not significantly different between high and low TCA_{M1} sessions (Figures 6G and 6H).

It appears puzzling that the proportions of IG and ET types in M1-projecting M2 neurons changed from low $TCA_{M2 \rightarrow M1}$ sessions to high $TCA_{M2 \rightarrow M1}$ sessions, whereas these proportions in M2 L2/3 did not change from low TCA_{M1} sessions to high TCA_{M1} sessions. When we imaged the same M2 L2/3 fields for seven training sessions after the IG trials were introduced, and examined the neurons that were pursued over these sessions (54.2 ± 10 neurons; $n = 5$ mice; Figure S9A), the TCA calculated from these pursued neurons was consistently high and the distribution of their trial-type selectivity was stable, although there were subsets of neurons with stable selectivity and others that flexibly changed selectivity (Figures S9B–S9G). In adjacent sessions, the trial-type selectivity of individual neurons was similar but frequently changed as the interval between sessions became longer (Figure S9H). In fact, the TCA calculated from the decoder that was modeled using the population activity within a session gradually decreased as the session progressed (Figure S9I). Thus, M2 L2/3 had stable neurons that were reliably active in each context and other neurons that flexibly changed their selectivity according to the trial type. IG and ET types of the M2 → M1 axons might originate from these flexibly changing neurons.

As TCA_{M1} increases, relationships between M2 and M1 change

In our previous study using a task with only ET trials, we showed that L2/3 population activities in M2 and M1 covaried across trials, which is not simply explained by trial-to-trial variability in behaviors (Terada et al., 2018). In the current IG-ET task, M2 L2/3 might also convey neuronal signals other than trial type and motor information to M1. As the interactions between brain regions are likely to be driven by the neuronal population (Perich et al., 2018; Semedo et al., 2019), we examined whether the influence of M2 population activity on the activity of individual M1 neurons changed as TCA_{M1} increased by adding the dimensionality-reduced population activity of M2 L2/3 to the behavior model (full model; see STAR Methods; Figure 7A). This addition increased R^2 (Figure 7B). To estimate the relative strength of the effect of M2 population activity on the activity in individual M1 neurons, we defined the difference in R^2 between the full

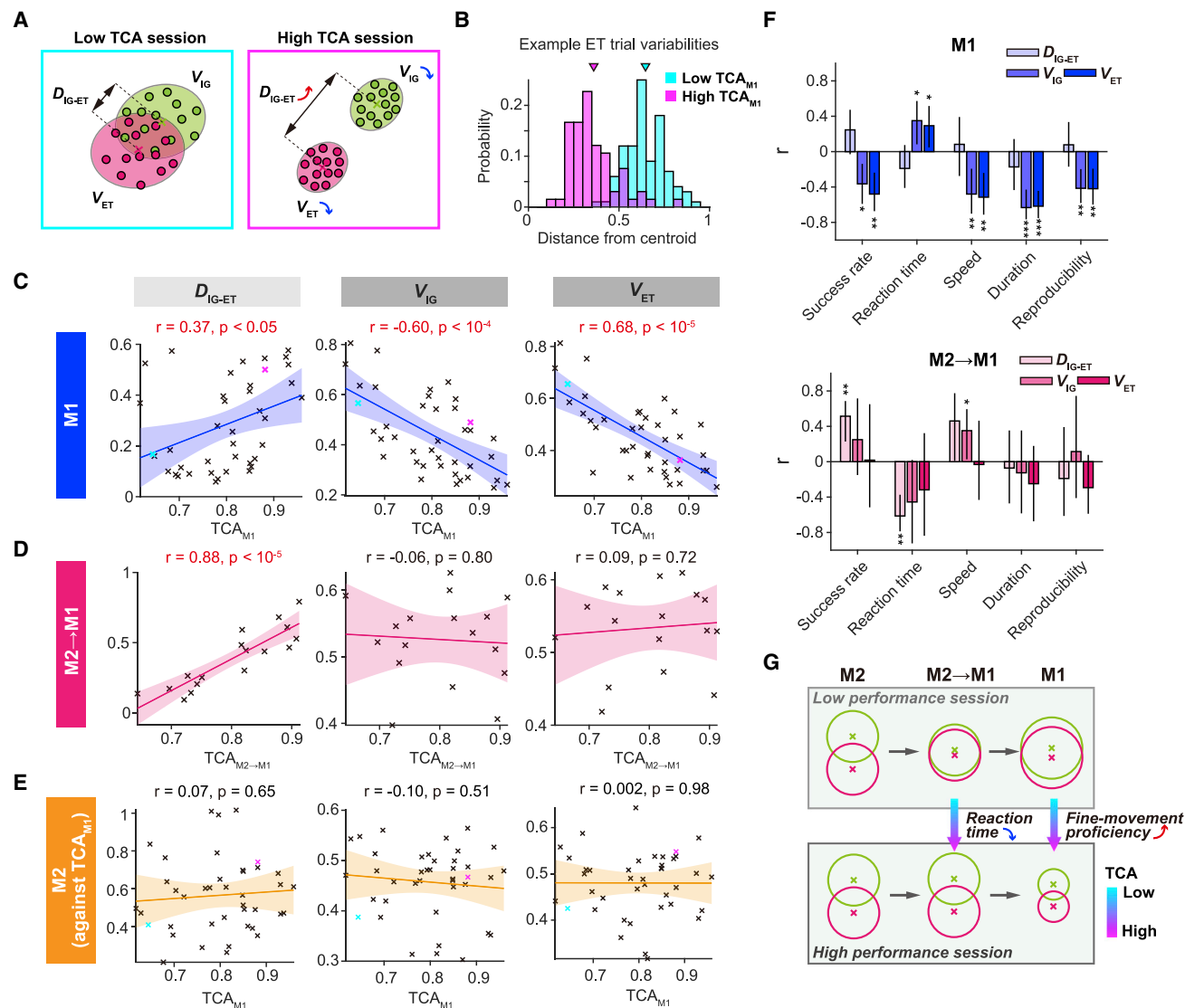


Figure 5. Distinct geometric changes in neuronal population activity associated with increases in TCA_{M1} and $TCA_{M2 \rightarrow M1}$

(A) Schematic illustration of geometric changes in the population activity from low to high TCA sessions. Each dot indicates the population activity of each trial in the hypothetical two-dimensional space. Green dots, IG trials; magenta dots, ET trials. Crosses indicate the centroids of the population activity in IG and ET trials. An increase in D_{IG-ET} , a decrease in V_{IG} , and/or a decrease in V_{ET} should increase TCA.

(B) Distributions of the trial-to-trial distance from the centroids in ET trials in the two sessions shown in Figure 4A. Inverted triangles indicate means.

(C) Scatterplots of D_{IG-ET} , V_{IG} , and V_{ET} against TCA_{M1} . Light-blue and magenta points correspond to the sessions shown in Figure 4A. r , Pearson's correlation coefficient; $n = 40$ sessions from six mice. Colored lines indicate the fit of linear models and shadows indicate 95% confidence intervals.

(D) Same plots as in (C) but for $TCA_{M2 \rightarrow M1}$. $n = 17$ sessions from four mice. Colored lines indicate the fit of linear models, and shadows indicate 95% confidence intervals.

(E) Scatterplots of the D_{IG-ET} , V_{IG} , and V_{ET} of M2 L2/3 neurons against TCA_{M1} . Sessions are the same as in (C) ($n = 40$ from six mice). Colored lines indicate the fit of linear models, and shading indicates 95% confidence intervals.

(F) Pearson's correlation coefficients between motor performance and D_{IG-ET} , V_{IG} , and V_{ET} . Error bars indicate 95% confidence intervals. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Pearson's correlation coefficient. $n = 40$ and 17 sessions from six and four mice for M1 L2/3 and M2 $\rightarrow$ M1 axons, respectively.

(G) Schematic illustration of geometric changes in the context-dependent population activity in M2 L2/3, M2 $\rightarrow$ M1 axons, and M1 L2/3 that were associated with improvement in motor performance. Crosses and circles indicate the centroids and variances of the population activity in each trial type, respectively. The cyan-magenta arrows indicate the increase in TCA in each structure. Shortening of the reaction time was also related to the emergence of context-dependent population activity in M1 L2/3.

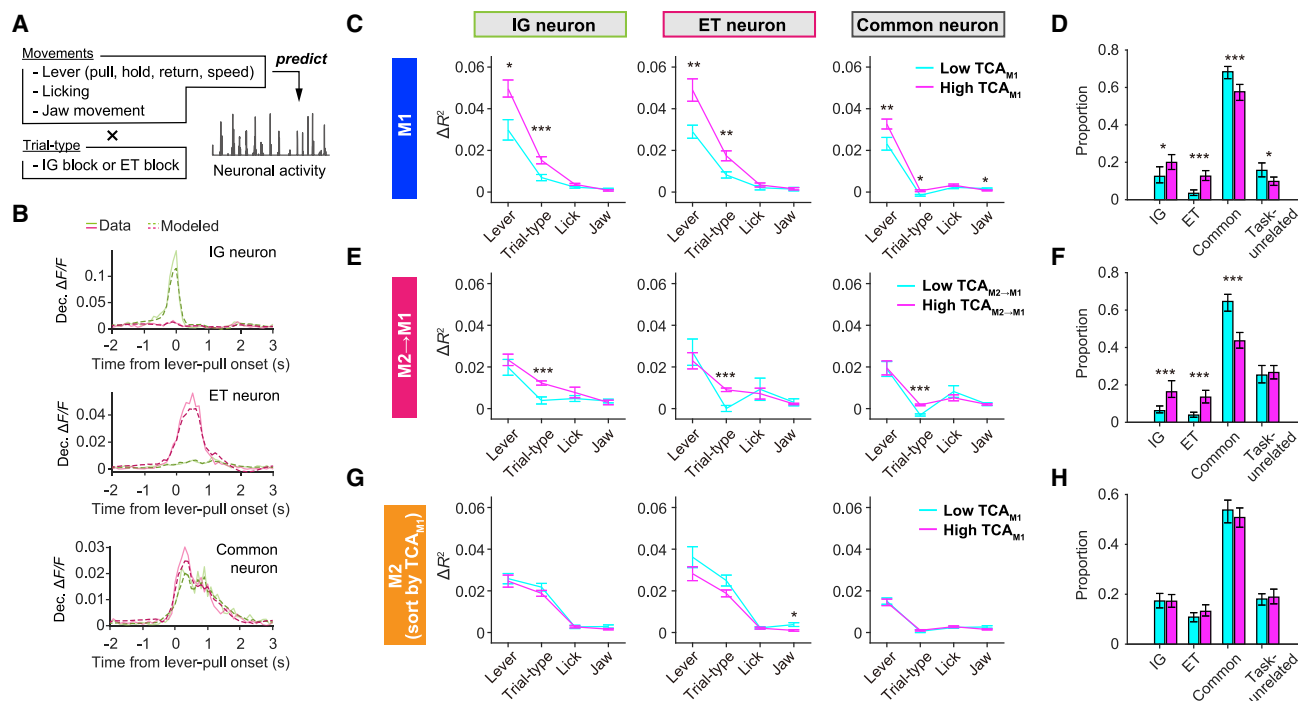


Figure 6. Trial-type and motor information carried by individual neurons

(A) Explanatory variables used to create the encoding model.
 (B) Examples of fittings by the encoding model to IG, ET, and common neurons. The actual deconvolved $\Delta F/F$ is shown by the solid line, and the prediction is shown by the dotted line. Trial averages are plotted for IG (green) and ET (magenta) trials.
 (C) Unique contributions of each type of variable in the IG, ET, and common neurons in M1 L2/3 in low (<0.8) and high (>0.8) TCA_{M1} sessions. ΔR^2 of each neuron type was averaged within each session. In sessions with low TCA_{M1} , only sessions with two or more neurons of each selectivity ($R^2 > 0.02$) were used in the analysis (18/19 and 13/19 sessions for IG and ET neurons, respectively). In common neurons, $n = 19$ and 21 sessions from five and four mice for low and high TCA_{M1} sessions, respectively. Error bars, SEM. Wilcoxon rank-sum test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
 (D) Proportions of M1 L2/3 neuron types in low and high TCA_{M1} sessions. Wilcoxon rank-sum test, * $p < 0.05$, *** $p < 0.001$, $n = 19$ and 21 sessions from five and four mice for low and high TCA_{M1} sessions, respectively. Error bars indicate 95% confidence intervals.
 (E) Same plots as in (C) but for M2 $\rightarrow$ M1 axons. Error bars, SEM. Wilcoxon rank-sum test, *** $p < 0.001$, $n = 7$ and 10 sessions from three and two mice for low (<0.8) and high (>0.8) $TCA_{M2 \rightarrow M1}$ sessions, respectively.
 (F) Proportions of M2 $\rightarrow$ M1 neuron types in low and high $TCA_{M2 \rightarrow M1}$ sessions. Wilcoxon rank-sum test, *** $p < 0.001$, $n = 7$ and 10 sessions from three and two mice for low and high $TCA_{M2 \rightarrow M1}$ sessions, respectively. Error bars indicate 95% confidence intervals.
 (G) Same plots as in (C) but for M2 L2/3 neurons against TCA_{M1} . Sessions were the same as in (C). Error bars, SEM. * $p < 0.05$, Wilcoxon rank-sum test, $n = 19$ and 21 sessions from five and four mice for low and high TCA_{M1} sessions, respectively.
 (H) Same plots as in (D) but for M2 L2/3 neurons. Wilcoxon rank-sum test, $n = 19$ and 21 sessions from five and four mice for low and high TCA_{M1} sessions, respectively. Error bars indicate 95% confidence intervals.
 See also [Figures S8](#) and [S9](#).

and behavior models divided by R^2 of the full model as the coupling index (Runyan et al., 2017). This coupling index decreased as TCA_{M1} increased (Figure 7C). This suggests that as TCA_{M1} increased, the contribution of M2 population activity to M1 activity reduced relative to the contribution of the behavioral variables including trial type. This is consistent with a previous study which showed that M1 ensemble activity related to behavior still remains even after M2 inactivation (Veuthy et al., 2020). When the coupling index calculated using the population activity of M2 IG and ET neurons minus the coupling index calculated using the M2 common neurons ($\Delta_{IGET-common}$) was plotted against TCA_{M1} , there was a slight but significant positive correlation (Figure 7D), although the proportions of these three types of M2 neurons were similar between high TCA_{M1} sessions and low TCA_{M1} sessions (Figure 6H). This suggests that as TCA_{M1}

increased, IG and ET neurons in M2 L2/3 were functionally closer to M1 L2/3 neurons than were M2 common neurons.

DISCUSSION

In this study, we clarified differences in the representation of behavioral contexts from a motor thalamic input in M2 to a pyramidal tract output in M1 during an IG and ET lever-pull task. The M2 L2/3 ensemble activity in the mouse showed high context dependency, as observed in previous studies in humans and nonhuman primates, which demonstrated that the higher-order motor cortex shows strong IG-type and ET-type activity (Okano and Tanji, 1987). This high context dependency was not due to a difference in a body movement between IG and ET trials. The context dependency of L5 PT neurons was consistently low,

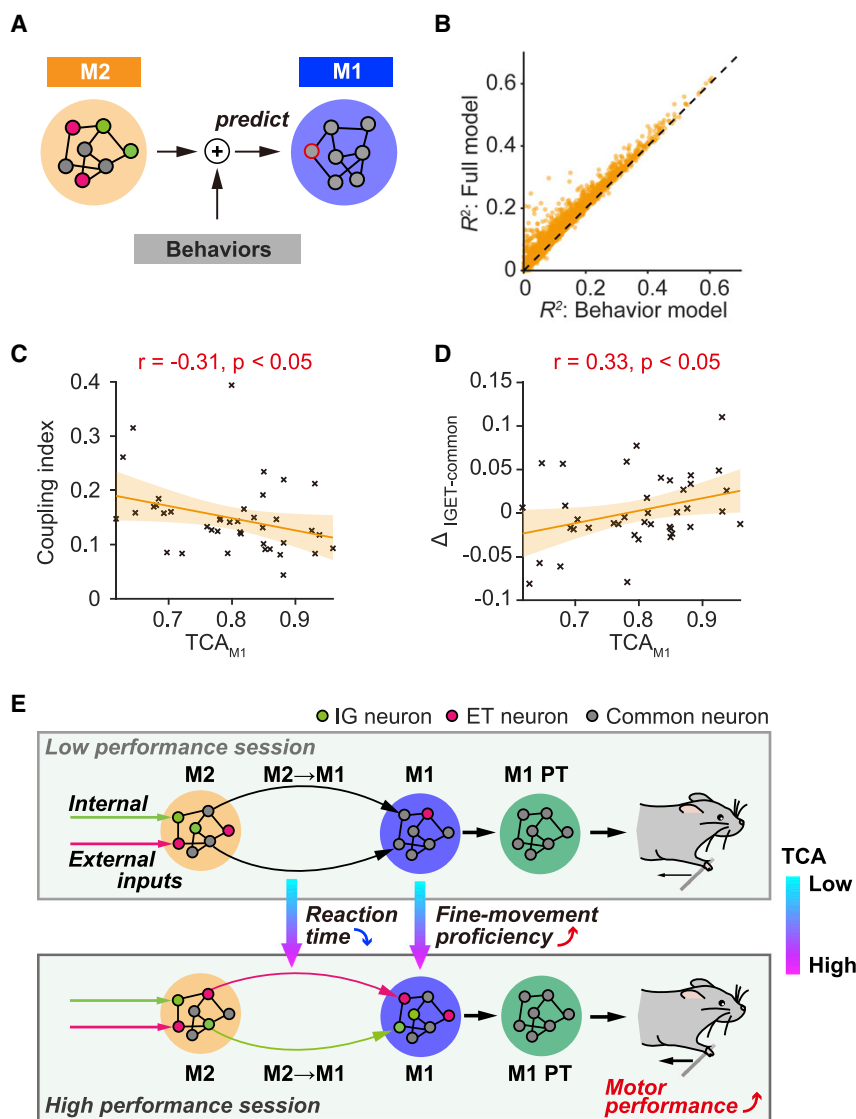


Figure 7. The influence of M2 population activity on individual M1 neurons

(A) Schematic illustration of the encoding model with neuronal population activity.

(B) Scatterplots of model performance (R^2) for the full model against the behavior model ($n = 5,183$ neurons in 40 sessions from six mice).

(C) Scatterplot of the coupling index against TCA_{M1} . Each dot indicates each session. The coupling index was averaged within each session. Pearson's correlation coefficient: $r = -0.31$, $p < 0.05$, $n = 40$ sessions from six mice. Colored line indicates the fit of the linear model, and shadow indicates 95% confidence interval.

(D) Scatterplot of the difference in the coupling index between the model with M2 population activity of IG and ET neurons, and the model with population activity of M2 common neurons ($\Delta_{IGET-Common}$) against TCA_{M1} . Pearson's correlation coefficient: $r = 0.33$, $p < 0.05$, $n = 40$ sessions from six mice. Colored line indicates the fit of the linear model and shadow indicates 95% confidence interval.

(E) Conceptual diagram of the context-motor transformation within the motor cortex for internally generated and external-cue-triggered movements. Dots that are colored green, magenta, and gray indicate IG, ET, and common neurons, respectively. The cyan-magenta arrows indicate the increase in TCA in each structure. The arrows that project from IG, ET, and common neurons in M2 L2/3 to IG, ET, and common neurons in M1 L2/3 are not directly demonstrated in the current study.

which is in accord with that reported for electrical recordings of M1 in the macaque (Okano and Tanji, 1987). We also found that TCA_{M1} varied between TCA_{M2} and the TCA of M1 pyramidal tract neurons, and that the variability of TCA_{M1} reflected the M1 L2/3 ensemble reorganization that was related to skilled motor performance. Furthermore, we clarified that MTh $\rightarrow$ M2 axons possessed higher TCA before lever-pull onset than the downstream motor cortex, and that M2 $\rightarrow$ M1 axons behaved like M1 L2/3 neurons with different relationships to the motor performance.

M2 L2/3 contained distinct subsets of neurons that exhibited stable IG-type and ET-type activity, as well as neurons that flexibly changed their type. The rodent M2 is responsible for a wide range of information processing, including action planning, decision making, rule shift, working memory, and behavioral adaptation (Barthas and Kwan, 2017; Chen et al., 2017; Murakami et al., 2014; Siniscalchi et al., 2016; Sul et al., 2011). Thus, M2 L2/3

plays crucial roles in both the stable reception of multiple types of information transferred from different areas and the flexible selection of the appropriate choice according to the context information. An obvious question arising is "which brain areas transmit IG-type and ET-type signals to M2?" The striatum generally modulates IG motor initiation (Klaus et al., 2019), and IG movements are more impaired than ET movements in patients with Parkinson's disease, with levodopa preferentially improving the performance of self-initiated movement in these patients (Burleigh-Jacobs et al., 1997; Jahanshahi et al., 1995). The high TCA of the MTh $\rightarrow$ M2 axons before the lever-pull onset might reflect IG-type signals that were generated in the striatum (Guo et al., 2017). In the initiation of ET movements, a signal from the pedunculopontine nucleus to M2 via several thalamic nuclei is necessary (Inagaki et al., 2022). This signal might convey the attendant sensory signal (Pan and Hyland, 2005). Neurons in the pedunculopontine nucleus also project to the VM (Inagaki et al., 2022). Thus, the MTh $\rightarrow$ M2 axons that we imaged might originate from VM neurons that received inputs from the basal ganglia and VM neurons that received inputs from the pedunculopontine nucleus, and they might possess similar proportions of IG and ET neurons (mean of 11.0% [CI 9.2%–14.1%] and 7.6% [CI 6.2%–9.2%], respectively; $n = 31$ sessions from six mice). The auditory signals to M2 in ET trials might also be

mediated by other pathways, including the striatum and cerebellum (Dacre et al., 2021; Gallero-Salas et al., 2021; Inagaki et al., 2022; Znamenskiy and Zador, 2013).

Subsets of M2 L2/3 neurons and/or M2 L5 neurons that changed their type might contribute to the emergence of M1-projecting IG and ET neurons. If M1-projecting M2 neurons of IG or ET type fired in synchrony with subsets of M1 neurons, synaptic plasticity might occur between them, and it is known that the formation rate of M2 → M1 axonal boutons changes during motor learning (Hasegawa et al., 2020). In visual and somatosensory cortices, the types of signals transferred are strongly related to their recipient area (Chen et al., 2013a; Glickfeld et al., 2013). However, M2 → M1 axons did not strongly encode the motor information that should be processed in M1. The optogenetic experiment suggests that M1, but not M2, was important for the lever-pull movement after the lever-pull onset. Thus, the common principle in well-tuned cortical processing from a donor area to its recipient area may be the selective transmission of the signal that is required for the initial condition in the recipient area. The major carriers of the context signal in the M2 → M1 pathway might be IG and ET neurons because the proportions of these neurons in M2 → M1 axons increased and $\Delta_{IG-ET-common}$ correlated with TCA_{M1} . The strong transmission of the context signal from M2 to M1 L2/3 may be related to the transition from the preparatory activity to the movement-related activity (Kaufman et al., 2016; Michaels et al., 2015). M1 also receives inputs from the VM and VA, and the ventrolateral nucleus (VL), which mainly receive projections from the cerebellum (Dacre et al., 2021; Tanaka et al., 2018). In a lever-pull task with only IG trials, VM inputs to M1 were related to the fine-movement proficiency, whereas VL inputs to M1 were related to success rate (Tanaka et al., 2018). In an ET lever-push task, VL inputs to M1 are critical for motor initiation (Dacre et al., 2021). Thus, M2 → M1 axons may play a role in the initiation of successful movement in cooperation with VL inputs to M1, while VM inputs to M1 may send important signals on the fine-movement proficiency.

Through the increase in context dependency in M2 → M1 axons, M1 L2/3 may be able to process context information in parallel with M2. However, in the population activity occurring around the lever-pull onset, the way that context signals were represented differed between M2 → M1 axons and M1 L2/3 neurons (Figure 5G). As $TCA_{M2 \rightarrow M1}$ increased, the D_{IG-ET} of M2 → M1 axons lengthened to ~ 0.6 , which was similar to the D_{IG-ET} of M2 L2/3, and V_{IG} and V_{ET} remained at the level of those of M2 L2/3 (~ 0.5) (Figures 5D and 5E). By contrast, as TCA_{M1} increased, the D_{IG-ET} of M1 L2/3 lengthened from ~ 0.2 to ~ 0.4 , which was shorter than the D_{IG-ET} of M2 L2/3, and both V_{IG} and V_{ET} of M1 L2/3 decreased from ~ 0.6 to ~ 0.3 , values that were less than those of M2 L2/3 (Figures 5C and 5E). The V_{IG} and V_{ET} of M2 L2/3 might reflect high susceptibility to trial-to-trial variability in sensory and nonsensory signals (Churchland et al., 2010; Goris et al., 2014). By contrast, M1 L2/3 could reduce the variability of signals not directly relevant to the forelimb movement. As the total decrease in the V_{IG} and V_{ET} (~ 0.6) was larger than the increase in the D_{IG-ET} (~ 0.2), the population activity space would become smaller still. Thus, at the neuronal population level, M1 L2/3 may compress the context-dependent ensembles into a smaller population activity space around the

lever-pull onset as motor performance improves (Figure 5G). However, this is a rather rough conjecture, and it needs to be clarified whether the space for the reorganized ensembles is selected within or outside of the space used before task performance improves (Gallego et al., 2017; Golub et al., 2018).

From the perspective of individual neurons, the relative contribution of the pathway of M2 IG and ET neurons → M1 IG and ET neurons → M1 PT neurons may be higher than the relative contribution of the pathway of M2 IG and ET neurons → M2 common neurons → M1 common neurons → M1 PT neurons as task performance increases (Figure 7E). In addition, lever-movement variables became strongly encoded in the IG and ET neurons in M1, and these neurons might therefore be able to effectively drive L5 PT neurons to output the motor command. These changes may effectively couple the processes of the context-dependent motor preparation, initiation, and execution. We propose a neuronal mechanism underlying the skillful execution of similar IG and ET movements as follows. The different IG and ET signals are first received by the stable context-dependent ensembles in M2 L2/3 neurons and are then transferred to newly formed context-dependent ensembles with weak motor encoding in M2 L2/3 and/or L5. A subset of the newly formed ensembles sends the context signals to M1 and activates the newly formed context-dependent ensembles in M1 L2/3. The activated M1 L2/3 ensembles generate the stable context-dependent motor-related signals that are relatively independent of M2 and drive the same subsets of L5 PT neurons in both contexts. The emergence of the context-dependent ensembles in the M2 → M1 pathway and M1 L2/3 in skilled motor performance may be a primitive form of the transition of different (e.g., output-null and output-potent) population activities for the control of more complex movements in the primate PM and M1 (Lara et al., 2018; Perich et al., 2018). M1 L2/3 ensembles reorganize in adaptation to sensory perturbations and motor learning (Heinrich et al., 2018; Huber et al., 2012; Masamizu et al., 2014; Omlor et al., 2019; Peters et al., 2014). Considering this, even at the expense of the cost of context-dependent ensemble formation in M1, M2-independent context-motor transformation within M1 may be effective for stable and accurate movements.

The pre-pull jaw movement was more frequently detected in ET trials than in IG trials, and this movement was regulated in the contralateral tJM1 (Mayrhofer et al., 2019; Mercer Lindsay et al., 2019). However, inactivation of the contralateral tJM1 did not inhibit the lever-pull performance or the jaw movement and licking for water consumption. The right tJM1 and/or trigeminal motor nucleus (Mercer Mercer Lindsay et al., 2019; Travers et al., 1997) were probably sufficient for the water consumption behavior. We speculate that the pre-pull jaw movement might reflect some internal state such as confidence and/or knowledge of the task structure, particularly its temporal structure. When comprehension of the task structure was high, it might have been maintained within each block so that the ET-type-dependent jaw movement occurred earlier than the ET-type-dependent M2 and M1 activity that was related to each lever-pull execution. In fact, the TCA at 2 s before the movement onset was slightly larger than 0.5 (Figures 3C and 3H). Thus, the task structure in each block might be monitored by a small subset of M2 and M1 neurons. Upstream of these neurons may be the medial prefrontal

cortex because this area is required for waiting for the action trigger stimulus, and its inactivation decreases M1 activity before the instructed movement (Narayanan and Laubach, 2006).

Limitations of the study

In the current study, we did not image the activity of axons from M2 to M1 layers other than L1. In addition, we did not image L5 neurons in M2. Thus, we did not exclude the possibility that M2 and M1 separately sent the initiation signal to the subcortical downstream areas and that these areas required both inputs. To demonstrate the flow of the context and motor information among the imaged areas, it will be necessary to transiently activate and inactivate each layer in M2 and M1, and the input axons from M2 and other areas to M1, to change the activity of the putative downstream neurons and behaviors. In addition, as the training of the IG task follows that of the ET task, this order might affect the reorganization of IG and ET neurons.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2022.111494>.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.-I.T. and M.M.; methodology, S.-I.T.; software, S.-I.T.; analysis, S.-I.T.; investigation, S.-I.T.; materials, K.K.; data curation, M.M.; writing – original draft, S.-I.T. and M.M.; writing – review and editing, S.-I.T. and M.M.; visualization, S.-I.T.; supervision, M.M.; project administration, M.M.; funding acquisition, S.-I.T. and M.M.

DECLARATION OF INTERESTS

S.-I.T. and M.M. are the authors of a patent on portions of super-wide-field two-photon microscopy that has been filed by the Japan Science and Technology Agency (JP6749648B2/CN107949800B/US11320639B2).

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV1-hSyn-R-CaMP1.07	Ohkura et al., 2012	N/A
AAV1-hSyn-GCaMP6s	Chen et al., 2013b	Addgene; 100843-AAV1
AAVretro-pmSyn1-EBFP-Cre	Madisen et al., 2015	Addgene; 51507-AAVrg
AAV1-hSyn-Flex-GCaMP6s	Chen et al., 2013b	Addgene; 100845-AAV1
AAV5-CaMKII-stGtACR1-FusionRed	Mahn et al., 2018	Addgene; 105679-AAV5
Chemicals, peptides, and recombinant proteins		
Muscimol	Sigma-Aldrich	M1523
Muscimol BODIPY TMR-X Conjugate	Thermo Fisher Scientific	M23400
Experimental models: Organisms/strains		
Mouse: C57BL/6JMsSlc	Japan SLC	RRID: MGI:5488963
Software and algorithms		
MATLAB R2020a	MathWorks	N/A
LabView 2018	National Instruments	N/A
ImageJ	National Institutes of Health	https://imagej.nih.gov/ij/index.html
Turboreg	Thévenaz et al., 1998	http://bigwww.epfl.ch/thevenaz/turboreg/
NoRMCorre	Pnevmatikakis and Giovannucci, 2017	https://github.com/flatironinstitute/NoRMCorre
CalmAn	Pnevmatikakis et al., 2016	https://github.com/flatironinstitute/CalmAn-MATLAB
DeepLabCut	Mathis et al., 2018	https://github.com/DeepLabCut/DeepLabCut
Code for analysis	This paper	https://doi.org/10.17632/5pfjk2kzjg.1

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Masanori Matsuzaki (mzakim@m.u-tokyo.ac.jp).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact without restriction. This study did not generate new unique reagents.

Data and code availability

All data reported in this study will be shared by the lead contact upon request. All original code has been deposited at <https://doi.org/10.17632/5pfjk2kzjg.1>, and is publicly available as of the date of publication. DOI is listed in the [key resources table](#). Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Animals

All animal experiments were approved by the Animal Experimental Committee of the University of Tokyo. C57BL/6 mice (male, aged 2–3 months, SLC, Shizuoka, Japan) were used for the experiments. All mice were provided with food and water *ad libitum* and housed in a 12:12 hour light-dark cycle (light cycle; 8 AM–8 PM).

METHOD DETAILS

Surgery for head plate attachment

Mice were anesthetized by intraperitoneal injection of a mixture of ketamine (74 mg/kg) and xylazine (10 mg/kg). After anesthesia, atropine sulfate (0.5 mg/kg) was administered intraperitoneally to improve breathing and bronchial secretion, and an eye ointment

(Tarivid, Santen Pharmaceutical, Osaka, Japan) was used to prevent eye-drying and infection. The head of the mouse was sterilized with 70% ethanol, and the skin covering the neocortex was incised. After the skull was exposed, a custom head plate (Tsukasa Giken, Sizuoka, Japan) (Hira et al., 2013) was attached to the skull using dental cement (Fuji lute BC, GC, Tokyo, Japan; and Bistite II or Estecem II, Tokuyama Dental, Tokyo, Japan). The surface of the intact skull was coated with dental adhesive resin cement (Super bond, Sun Medical, Siga, Japan) to prevent drying and inflammation. The number of mice per cage was 2–5 before the head plate was attached, but the mice were single-caged after attachment to avoid damage to the head plate and glass window. The anti-inflammatory analgesic carprofen (6 mg/kg; Remadile, Zoetis, NJ, USA) was injected intraperitoneally after all surgical procedures. After the attachment, a recovery period of at least 1 day was allowed before any other procedure.

Virus production

For imaging of R-CaMP1.07, GCaMP3 DNA of pAAV-human synapsin I promoter (hSyn)-GCaMP3-WPRE-hGH polyA (Masamizu et al., 2014) was replaced with R-CaMP1.07 DNA from pN1-R-CaMP1.07 vector construct (Ohkura et al., 2012). AAV1-hSyn-R-CaMP1.07 (titer $\sim 1.3 \times 10^{13}$ GC/mL) was produced with pAAV2-1 and purified as described previously (Kobayashi et al., 2016).

Surgery for *in vivo* imaging

One hour before surgery, dexamethasone sodium phosphate (1.32 mg/kg), the antibiotics sulfadiazine (24 mg/kg) and trimethoprim (4.8 mg/kg), and the anti-inflammatory analgesic carprofen (6 mg/kg) were administered intraperitoneally. Virus injections were performed with a pulled quartz pipette (broken and beveled to an outer diameter of 25–30 μ m; Q100-50-10, Sutter Instruments, CA, USA) and a 5 μ L Hamilton syringe (87930, Hamilton, NV, USA). The pipette was back-filled with mineral oil (23306-84, Nacalai Tesque, Kyoto, Japan) and front-loaded with virus solution. Isoflurane (1–1.4%) inhalation was used for anesthesia throughout surgery.

For fast sequential imaging of M2 and M1 (M2-M1 imaging), a rectangular craniotomy of about 2 $\times$ 4 mm centered at 1.0 mm anterior to the bregma and 1.2 mm left of the midline was made. After the brain surface was exposed, AAV1-hSyn-R-CaMP1.07 was diluted 1:5 in saline and injected into the left M2 and M1 (M2 centered at 2.5 mm anterior and 0.8 mm lateral to the bregma, and 300 μ m depth from the brain surface; M1 centered at 0.5 mm anterior and 1.2 mm lateral to the bregma, and 300 μ m depth from the brain surface; 120 nL per site; two or three sites per region). For the imaging of the same L2/3 fields in M2 (Figure S9), a circular craniotomy of about 2 mm in diameter was performed above the left M2. AAV1-hSyn-R-CaMP1.07 was diluted 1:5 in saline and injected into M2 (300 μ m depth from the brain surface; 120 nL per site; two or three sites). For the M2 $\rightarrow$ M1 axon imaging, a small craniotomy above the left M2 and a circular craniotomy of about 2 mm diameter above the left M1 were performed. AAV1-hSyn-GCaMP6s (titer $\sim 3 \times 10^{13}$ GC/mL, obtained from Addgene, MA, USA) (Chen et al., 2013b) was diluted 1:5 in saline and injected into the left M2 (centered at 2.5 mm anterior and 0.8 mm lateral to the bregma, and 500 μ m depth from the brain surface; 200 nL). For the MTh $\rightarrow$ M2 axon imaging, a small craniotomy above the left motor thalamus (centered at 1.2 mm posterior and 1.0 mm lateral to the bregma) and a circular craniotomy of about 2 mm diameter above the left M2 were performed. AAV1-hSyn-GCaMP6s was injected into the left motor thalamus (centered at 1.2 mm posterior and 1.0 mm lateral to the bregma, and 3.75 mm depth from the brain surface; 100 nL) without dilution. GCaMP-expressing sites included mainly VM, VA, and the ventral lateral nucleus (VL; Figure S4A). Although VL mainly receives output from the deep cerebellar nucleus, VL neurons project their axons to the deep layer, but not L1 (Tanaka et al., 2018). Thus, we considered the imaged L1 axons to originate mainly from VM and VA. For the M1 PT neuron imaging, a small craniotomy above the left Pons (centered at 3.8 mm posterior and 0.5 mm lateral to the bregma) and a circular craniotomy of about 2 mm diameter above the left M1 were performed. AAVretro-pmSyn1-EBFP-Cre (titer $\sim 3 \times 10^{12}$ GC/mL, obtained from Addgene) (Madisen et al., 2015) and AAV1-hSyn-Flex-GCaMP6s (titer $\sim 3 \times 10^{13}$ GC/mL, obtained from Addgene, diluted 1:5 in saline) were injected into the left Pons (centered at 3.8 mm posterior and 0.5 mm lateral to the bregma, and 4.8, 5.0, and 5.2 mm depths from the brain surface; 70 nL per depth) and left M1 (centered at 0.5 mm anterior and 1.2 mm lateral to the bregma, and 850 μ m depth from the brain surface; 200 nL), respectively. For the S1 imaging, a circular craniotomy of about 2 mm diameter was performed above the left forelimb-related S1 (Levy et al., 2020; Prsa et al., 2019). AAV1-hSyn-R-CaMP1.07 was diluted 1:5 in saline and injected into S1 (centered at 0.0 mm anterior and 2.5 mm lateral to the bregma, and 300 μ m depth from the brain surface; 120 nL per site; two or three sites). The dura was left intact as far as possible, although in some cases it was thick and opaque above M2, and was therefore carefully removed.

After the injection, the pipette was withdrawn, a double-layer glass window was pressed onto the brain surface, and the edge was sealed with cyanoacrylate glue (Vetbond, 3M, MN, USA) and dental adhesive resin cement (Super bond). The double-layer glass window was constructed by using a UV-curing optical adhesive to join a 3 $\times$ 5 mm rectangular glass coverslip (No. 0 thickness; Matsunami Glass, Osaka, Japan) and a 2 $\times$ 4 mm rectangular glass coverslip (No. 4 thickness) for M2-M1 imaging, or a 3 mm diameter glass coverslip (No. 0 thickness) and a 1.5 mm diameter glass coverslip (No. 5 thickness) for other imaging windows. Mice were then returned to their home cages. The surgery was performed after the ET task training finished, except for the chronic imaging of the same L2/3 fields in M2 (Figure S9). In this case, the surgery was performed 1 day after the head plate attachment to obtain a sufficient expression period. Before the imaging experiments were conducted, the mice were allowed to express sufficient quantities of transgenes for 2–3 weeks.

Behavioral task

Before starting behavioral training, the mice were water-deprived in their home cages and were maintained at 80–85% of their normal weight throughout the experiments. If body weight had not sufficiently increased after behavioral training, an appropriate volume of

water or agar gel was given. Mice were head-fixed via their head-plates and inserted into a body chamber. The sessions were carried out once per day, with a duration of 30–60 min. First, we trained the mice to perform external-cue triggered lever-pull (ET) trials (Terada et al., 2018). In these ET trials, the mice were trained to use their right forelimbs to pull the lever ~ 3 mm for 400 ms within 500 ms of the onset of a 100 ms sound cue (11 kHz) to obtain a 4 μ L drop of water delivered from a spout near their mouth. The reward was given at 400 ms after the initiation of the successful lever-pull. As there was a delay of 176.1 ± 1.0 ms (mean $\pm$ SD, $n = 20$ trials) between the controller's command and the actual arrival of the reward at the spout tip, the time including this delay is shown as the time of the reward delivery. The lever-pull threshold was set at about 20% of the maximum displacement. The sound cue was delivered 2.5–3.5 s after the lever was returned in the preceding lever-pull. To reduce the frequency of lever-pulls in the intertrial interval, if the lever was pulled before the next sound cue, the waiting time was reset so that the next sound cue was delivered 2.5–3.5 s after the lever was returned.

The training proceeded as follows. In the first and second sessions, the reward was delivered when the mice licked the spout within 1 s of the end of the tone after the cue was presented, irrespective of any lever movement. Next, the forelimb lever-pull training was started, with the duration of lever-pull being extended from 1 ms to 200 ms, and then finally to 400 ms during the training sessions. The mice were trained over 10–14 sessions to finally pull the lever for 400 ms with a high (>80%) success rate. Then, internally generated lever-pull (IG) trials (Hira et al., 2013; Masamizu et al., 2014; Tanaka et al., 2018) were introduced to the ET trials. In these IG trials, the mice had to wait without pulling the lever for at least 2.6 s and then pull the lever for 400 ms to obtain the reward. No sound cue was delivered. If the lever was pulled without waiting for 2.6 s, or the pull duration was less than 400 ms, the wait period started again after the lever was returned. The blocks of the IG and ET trials (IG and ET blocks, respectively) were switched every 3 minutes during each session. This was different from a previous rat experiment in which IG and ET trials were randomly switched in each trial (Kimura et al., 2017). Each session started with the ET block. The mice were trained on this task for 10–18 sessions before moving on to the imaging sessions. During the imaging sessions, the number of successful trials per minute was 5.7 ± 0.20 in the IG block and 7.8 ± 0.28 in the ET block. The success rate in the ET trials was 0.861 ± 0.0002 . The reaction time in successful ET trials was 131.6 ± 8.3 ms ($n = 129$ sessions from 28 mice). A program written with LabVIEW (National Instruments, TX, USA) was used to control the task devices and record the lever trajectory, sound cue timing, reward timing, frame triggers of the two-photon imaging, and high-speed video cameras.

Design of the micro-opto-mechanical device

The design of the micro-opto-mechanical device was slightly modified from that previously described (Terada et al., 2018) to increase the rotation speed. Briefly, the micro-opto-mechanical device consisted of four parts: a pair of mirrors, a mirror holder, a rotation mount, and a pillar. We changed the transmission mechanism of the rotation mount from gears to a timing belt and pulleys. A large-diameter custom-made timing pulley (2GT belt type, 90 teeth) was installed on the mirror holder, a small-diameter timing pulley (P34-2GT-BLP-4C-5, NBK, Gifu, Japan) was installed on a stepping motor, and a 4-mm-wide timing belt (GBN2642GT-40, Misumi, Tokyo, Japan) was installed to transmit power between the two pulleys. To dampen the transmission of vibration generated by the device, six vibration isolation rubbers (RB-25, Kurashiki Kako, Okayama, Japan) were inserted between the base plate of the pillar and the stage where the microscope was installed. An acceleration sensor (MMA7361, Freescale Semiconductor, TX, USA) was used to evaluate the vibration.

Two-photon imaging

Imaging was conducted with a two-photon microscope (FVMPE-RS, Olympus, Tokyo, Japan) with a broadly tunable ultrafast laser (InSight DS + Dual, Spectra Physics, CA, USA). The wavelength of the laser was set at 940 nm for GCaMP6s imaging and 1100 nm for R-CaMP1.07 imaging. A long-working distance (WD) high-numerical aperture (NA) objective (XLPLN10XSVM, 10 $\times$, WD, 8.0 mm, NA, 0.6; Olympus) was used for M2-M1 imaging, and a short-WD high-NA objective (XLPLN25XWMP2, 25 $\times$; WD, 2 mm; NA, 1.05; Olympus) was used for the other imaging. Fluorescence emission was collected with a GaAsP photomultiplier tube (Hamamatsu Photonics, Shizuoka, Japan) after separation from the excitation beam path through a 690 nm shortpass dichroic mirror, a 570 nm dichroic mirror, and a bandpass filter of 495–540 nm for GCaMP6s or a bandpass filter of 575–645 nm for R-CaMP1.07 (FV30-FGR, Olympus). A piezo objective positioner (P-725KHDS, Physik Instrumente, Karlsruhe, Germany) controlled by microscope control software (FV30S-SW, Olympus) was used to rapidly change the imaging field along the optical axis. Before imaging, a two-axis goniometer attached to the head-fixing device was adjusted so that the cranial window and the objective lens were nearly horizontal. For M2-M1 imaging, the micro-opto-mechanical device was placed between the objective lens and the window to image L2/3 of M2 and M1. The procedures for the setting and imaging were previously described in more detail (Terada et al., 2018). A series of 14,500 continuous images were acquired at 10 Hz. The size of each field of view (FOV) was 372.9×636.4 μ m (300 $\times$ 512 pixels). All focal planes were located 110–200 μ m below the cortical surface. For the tracking of the same L2/3 neurons in M2, standard two-photon imaging was performed in a single FOV using the 25 $\times$ objective lens without the micro-optomechanical devices. A series of 44,000 continuous images were acquired at 30 Hz. The size of each FOV was 509.1×509.1 μ m (512 $\times$ 512 pixels). All focal planes were located 150–180 μ m below the cortical surface. Imaging of S1 was also conducted using the imaging parameters described above. For imaging of axonal boutons, the back aperture of the 25 $\times$ objective lens was underfilled with the diameter-shortened (7.2 mm) laser beam, to reduce the effective excitation NA of the objective (Kondo et al., 2017) and the effect of motion along the optical axis on image quality (Yoshida et al., 2018). A series of 44,000 continuous images were acquired at 30 Hz. The size of each FOV

was $101.8 \times 101.8 \mu\text{m}$ (512×512 pixels). All focal planes were located 10–50 μm below the cortical surface to image the axons projecting to L1 in M1 or M2. In M2 $\rightarrow$ M1 axon imaging, subsets of the L1-projecting axons should originate from L2/3 neurons in M2 (Hasegawa et al., 2020). Therefore, we considered that these L1 axons were suitable for examining the trial-type information transmitted from M2 to M1. For imaging of the apical dendrites of M1 PT neurons, the imaging was conducted with the underfilled objective, similar to the imaging of axonal boutons. A series of 44,000 continuous images were acquired at 30 Hz. The size of each FOV was $363.7 \times 363.7 \mu\text{m}$ (512×512 pixels). All focal planes were located 400–500 μm below the cortical surface to image apical dendrites before they branched (or as much as possible before branching). Imaging was performed in one field in each session. If the number of successful trials in the imaging session was less than 40 in each IG trial and ET trial, or if the success rate of the ET trials was less than 0.4, the sessions were excluded from the analysis, except for the chronic imaging of the same L2/3 neurons.

Video monitoring

Two high-speed video cameras (Scout scA640-70gm, Basler, Germany, with an M0814-MP fixed-focus lens of 8 mm focal length, Computer, NC) were used to capture the upper parts of the bodies and faces of the mice. The cameras were synchronized with a pulse signal generated by the data acquisition board used for the task acquisition, and the images were acquired at 70 Hz. An infrared LED light source (940 nm, 60 LEDs) was positioned at the back of each camera to illuminate the video monitoring areas without affecting the two-photon imaging.

Pharmacological inactivation

In the M2 or M1 suppression experiments, three mice were used for the muscimol injection after they completed the IG-ET task training sessions. By the day before the injections, 1 mm diameter craniotomies were made above the left M2 (2.5 mm anterior and 1 mm lateral to bregma) and left M1 (0.5 mm anterior and 1.2 mm lateral to bregma) under isoflurane anesthesia. After the craniotomies were made, they were sealed with Kwik-Cast (WPI, FL, USA). In the inactivation session, a small volume of muscimol (80 nL; 5 $\mu\text{g}/\mu\text{L}$; M1523, Sigma-Aldrich, MO, USA; at a speed of 40 nL/min) was injected into M2 or M1 at 500 μm depth from the brain surface with a beveled pipette (25–30- μm -diameter tip) under isoflurane anesthesia. Five minutes after the start of the injection, the glass capillary was carefully withdrawn. The mice were then kept in their home cages for 30–60 minutes before the start of the test sessions to recover from the anesthesia. In the control sessions, saline was injected into M2 and M1. Muscimol and saline were alternated from day to day. To evaluate the effect of muscimol on task performance, we quantified the number of rewards and the number of pull initiations in IG trials, and the number of rewards, the reaction time, and success rate in ET trials, during the first 24 minutes of the session. The number of pull initiations was defined as the number of lever-pulls that continuously exceeded the lever-pull threshold for more than 100 ms in the IG block. In the ET trials, the time between the cue-onset and the lever trajectory first crossing the lever-pull threshold was defined as the reaction time. The success rate was defined as the number of rewards in ET trials divided by the number of cue presentations. The forelimb position accuracy was calculated from videos acquired with a high-speed camera (see ‘Video monitoring’ and ‘Video tracking’ sections in STAR Methods). It was defined as the proportion of trials in which the right forelimb was on the lever at the time of cue presentation (the position of the right forelimb was <25 pixels vertically from the lever). In the tJM1 inactivation experiment, a small volume of muscimol was injected into the left tJM1 (1.8 mm anterior and 2.5 mm lateral to the bregma) in the post-training mice ($n = 4$ mice) under the same conditions as used for M2 and M1.

Quantification of the spread of muscimol

To evaluate the spread of muscimol, a small volume of muscimol BODIPY TMR-X conjugate (80 nL; 2.67 $\mu\text{g}/\mu\text{L}$; M2400, ThermoFisher Scientific, MA, USA; injected at a speed of 40 nL/min) was injected into M2 or M1 under the same conditions as used in the pharmacological inactivation experiment. One hour later, the mice were deeply anesthetized by intraperitoneal injection of a mixture of ketamine and xylazine, and were then perfused transcardially with PBS followed by a solution of 4% paraformaldehyde (09154-85, Nacalai Tesque). The brains were removed and stored in the fixative overnight. The fixed brains were observed under a fluorescence microscope (Axio Zoom.V16; Carl Zeiss, Jena, Germany) equipped with a $1\times$ objective (Plan-NEOFLUAR Z 1.0/0.25; Carl Zeiss). The fluorescent dye was excited with a fluorescence light source (HXP 200 C; Carl Zeiss) and a filter set (63HE; Carl Zeiss; 572/25 nm excitation filter, 590 nm dichroic mirror, and 629/62 nm emission filter), and the emitted fluorescence was collected with a scientific CMOS camera (Sona 4.2B-11; Andor Technology, Belfast, UK). A filter set was used to obtain the shape of the entire brain (38HE; Carl Zeiss; 470/40 nm excitation filter, 495 nm dichroic mirror, and 525/50 nm emission filter). Fluorescence images of brains injected with M2 or M1 were aligned using the shape images so that the two brains overlapped, and the points of maximum fluorescence intensity in each image were defined as the center of injection of M2 or M1, respectively. A fluorescence intensity profile along a straight line passing through both centers was then quantified in each image. Since the molecular weight and hydrophilicity of the muscimol BODIPY (607) and muscimol (114) differed, and these were diffused in the intercellular space in the cerebral cortex, the spread of muscimol could not be accurately estimated in this experiment. However, given that the diffusion constant in aqueous solution is inversely proportional to the square root of the molecular weight, and that the diffusion distance is proportional to the square root of the diffusion constant, we supposed that the spread of muscimol was at most less than twice that of muscimol BODIPY, and would therefore still be less than 2.5 mm. In addition, muscimol injection to tJM1, which is spatially separated from the forelimb M2 and M1 that we injected at a distance of more than 1.0 mm, did not cause any apparent deterioration in the lever-pull movement.

Light illumination of stGtACR1-expressing neurons

We injected AAV5-CaMKII-stGtACR1-FusionRed (titer $\sim 4.5 \times 10^{12}$ GC/mL, obtained from Addgene) into the left M2 (300 and 800 μm depth from the brain surface; 120 nL for each depth) and the left M1 (300 and 800 μm depth from the brain surface; 120 nL for each depth) in mice attached by head plate ($n = 5$ mice). A glass window (2×4 mm rectangular) was placed over the injection area in the same way as for M2-M1 imaging. After the surgery, mice were trained in the IG-ET task. Activation of stGtACR1 was performed by irradiating the target area with a 594 nm laser (OBIS LS 594 nm 100 mW; Coherent, CA, USA). This wavelength was selected because it has higher transmittance within brain tissues than blue or green light (Mahn et al., 2018), and was expected to effectively suppress the activity of excitatory neurons located deep in the tissue, such as at L5. The laser beam was targeted onto the brain surface through a microscope equipped with a two-dimensional galvanometric scanning system, as described previously (Kondo and Matsuzaki, 2021). In brief, the collimated laser beam emitted from the light source was coupled into a multimode optical fiber (M50L02S-A, $\varnothing 50$ mm, 0.22 NA; Thorlabs, NJ, USA) with a fiber coupler (PAF2S-2A, $f = 2.0$ mm; Thorlabs). The laser that passed through the optical fiber was recollimated using a fiber coupler (PAF2S-7A, $f = 7.5$ mm; Thorlabs) and entered into a two-dimensional galvanometric scanning system (FluoView; Olympus). The scanning laser beam was focused on the brain surface through an achromatic doublet lens (AC254-200-A-ML, $f = 200$ mm; Thorlabs). The beam shape under the focusing lens (Figures S1F and S1G) was measured using a beam analyzer (Super BeamAnalyzer; Melles Griot, CA, USA). The intensity of the laser was controlled with a laser controller (OBIS LX/LS single laser remote; Coherent) and was set to 3 mW after the focusing lens.

The light was illuminated onto the left M2 (2.5 mm anterior and 0.8 mm lateral to the bregma as the center of the light spot) or the left M1 (0.5 mm anterior and 1.2 mm lateral to the bregma). Continuous light stimulation was applied for 1 s from the moment the lever crossed the lever-pull threshold. To restrict the stimuli to lever-pulls that could be successful lever-pulls, we used the lever-pull trials with the waiting time more than 2.6 s in the IG trials and the first lever-pull after the cue-onset in the ET trials. Illumination was applied or not with a probability of 0.5, light stimulation to the brain was performed with a probability of 0.25, and the head plate was illuminated with a probability of 0.25. The head-plate stimulation was to mimic the scattering of the light when the brain was stimulated. To prevent the mice from recognizing the timing of the light stimulus, masking LEDs (orange, blinking at 20 Hz) were placed in front of their eyes to keep them blinking throughout the session. To obtain a sufficient number of trials for behavioral analysis, only one brain stimulation site was used per session.

Quantification of the effect of the light illumination was performed as follows. The pull distance was defined as the average position of the lever from 0 to 0.4 s from the light stimulus onset. The latency of the displacement of the lever trajectory was defined as the time from the illumination onset to the time point when the trajectory started to differ from the trajectory in the head-plate stimulation for more than 20 ms ($p < 0.05$, Wilcoxon rank sum test). Only sessions in which the start of the trajectory change was detected within 400 ms after the illumination onset were used in the latency analysis ($n = 8$ sessions from four mice).

Histology

At the end of the imaging and optogenetic experiments for M2 $\rightarrow$ M1 axons, MTh $\rightarrow$ M2 axons, and M1 PT neurons, the mice were deeply anesthetized by intraperitoneal injection with a mixture of ketamine and xylazine, and then perfused transcardially with PBS followed by a solution of 4% paraformaldehyde. Their brains were removed and stored in the fixative overnight. Coronal or sagittal sections were made at 100 μm thickness. DAPI (1:1000; D8417, Sigma-Aldrich) or NeuroTrace Blue (1:200, N21479, Invitrogen) was used to visualize the cytoarchitecture. Epifluorescence images were acquired using a fluorescence microscope (IX83, Olympus) equipped with a $4\times$ or $10\times$ objective lens and fluorescence filter sets appropriate for DAPI, GFP, and FusionRed.

GCaMP6s-expressing areas are determined according to the atlas (Franklin and Paxinos, 2013). In AAV injection for MTh $\rightarrow$ M2 axon imaging, apart from VA, VL, and VM, parts of the anteromedial nucleus, centrolateral nucleus, and mediodorsal nucleus were also labeled (Figure S4A). The anteromedial nucleus is included in the motor thalamus in Hooks et al. (Hooks et al., 2013). According to Unzai et al. (Unzai et al., 2017), the centrolateral nucleus mainly projects to layer 6. The projection of the mediodorsal nucleus neurons to M2 layer 1 is anatomically very weak (Kuramoto et al., 2017). The VA and VM strongly project their axons to L1 of M2, as well as L1 of M1 (Kuramoto et al., 2009, 2015). Thus, we considered that the imaged axons originated mainly from the motor thalamus.

Image processing for two-photon calcium imaging data

Analyses were performed using MATLAB (R2020a; Mathworks, Natick, MA, USA) and ImageJ (National Institute of Health, Bethesda, MD, USA). For the imaging of neuronal somata, lateral displacements of acquired image sequences were corrected using the NoRMCorre algorithm (Pnevmatikakis and Giovannucci, 2017) with the rigid motion correction option. For imaging of axons and dendrites, the displacements were first corrected using Turboreg (Thévenaz et al., 1998), and then using the NoRMCorre algorithm with the non-rigid motion correction option. After the motion correction, temporal averaging over two frames (excluding M2-M1 imaging) and spatial averaging to halve the image size were performed to reduce computational cost. We refer to the image after the above procedures as I_{raw} . A constrained non-negative matrix factorization algorithm (CalmAn package for MATLAB, <https://github.com/flatironinstitute/CalmAn-MATLAB>) (Pnevmatikakis et al., 2016) was used to extract neuronal activities from I_{raw} images. The algorithm decomposes the I_{raw} into a time course of inferred calcium transients (temporal component) and their spatial position and shape (spatial component; ROI), together with components that model the background/neuropil activity (I_{back}). After the identification of these components, we multiplied the spatial components and the background-subtracted raw image ($I_{\text{raw}} - I_{\text{back}}$) at each pixel,

then averaged over the pixels within each ROI to determine the fluorescence signals for each ROI. Then, detrended relative fluorescence changes ($\Delta F/F$) were calculated for eight percentile values over an interval of ± 30 s around each sample time point (Dombeck et al., 2007). The $\Delta F/F$ of each ROI was then deconvolved using the constrained AR-2 OASIS algorithm (Friedrich et al., 2017) with decay constants optimized separately for each ROI; we refer to these as deconvolved $\Delta F/F$. The deconvolved $\Delta F/F$ were resampled to 100 Hz using linear interpolation (MATLAB's *interp1* function), and synchronized with the behavioral data and frame triggers of the high-speed video and microscopy for later analysis. For the imaging of neuronal somata, pseudo neuropil structures (apical dendrites and/or axons) in L2/3 planes were removed by excluding those ROIs with a size smaller than $50 \mu\text{m}^2$ or oblateness greater than 0.5 (Terada et al., 2018). To track the same L2/3 neurons (Figure S9), we used an algorithm for alignment of ROIs across different sessions ('register_ROIs', provided as part of the CalmAn package) (Giovannucci et al., 2019). ROIs were mapped in each pair of sessions (Session 1 and one of Sessions 2–7) and those ROIs that were identified in all six pairs were considered as the pursued neurons and used for the analysis. In the axon and dendrite imaging data, those ROIs showing high correlation in their activity ($r > 0.7$) were considered to be from the same origin. The signal-to-noise ratio for each of these correlative ROIs was defined as the product of the maximum values of the temporal and spatial components, following the methods of a previous study (Pnevmatikakis et al., 2016), and the ROI with the highest signal-to-noise ratio was selected as a representative ROI.

In the current study, we used RCaMP1.07 for imaging of neuronal somata and GCaMP6s for imaging of axonal boutons and apical dendrites. Although it was reported that RCaMP1.07 has single spike sensitivity in imaging of neuronal somata (Bethge et al., 2017), its sensitivity might be lower when used for imaging of axonal boutons and dendrites. In addition, it has not been reported whether the RCaMP1.07 response in axonal boutons and dendrites corresponds to firing. By contrast, previous studies involving our group showed that the GCaMP6f response in axonal boutons corresponds to firing in axons (Broussard et al., 2018; Tanaka et al., 2018). In addition, GCaMP6f has already been used to image apical dendrites (Peters et al., 2017). GCaMP6s is more sensitive than GCaMP6f, and we therefore used GCaMP6s to increase the sensitivity for axonal boutons and dendrites. We confirmed that GCaMP6s, similar to GCaMP6f, can detect activity in dendrites in a similar manner as in neuronal somata (Figures S4I and S4J). To estimate the possible effects of differential sensitivity of the calcium indicator on TCA, we artificially degraded the data from M2 L2/3 and M1 L2/3 neurons by removing small calcium responses (Figures S4P and S4Q). In this analysis, calcium events from deconvolved $\Delta F/F$ were identified by a peak detection algorithm (MATLAB's *findpeaks* function; minimum peak prominence, 0.05). From the distribution of the peak values, 10, 20, 30, 40, and 50 percentile values were set as thresholds, and time points below the thresholds were set to 0 to simulate decreased sensitivity to small calcium events.

Video tracking

DeepLabCut (<https://github.com/DeepLabCut/DeepLabCut>) (Mathis et al., 2018) was used to automatically track forelimb and orofacial movements in the high-speed videos obtained. To make a training dataset, 150 representative frames were extracted from the videos acquired during M2-M1 imaging using the k-means algorithm for each camera (25 frames per session, one session per animal). Then, for the face side images, the tip of the nose, tip of the tongue, upper lip, lower lip, jaw, and tip of the spout were manually labeled. For the forelimb side images, the tip of the lever, thumb, index finger, middle finger, ring finger, and little finger were manually labeled. We labeled all fingers to increase the stability of the model predictions, but only the middle finger was used for the forelimb position in the analysis. Using the labeled images, we trained a network model with 500 000 iterations and confirmed convergence of the loss function (<0.001). This model was applied to all data to predict the position of each body part. The predicted positions were resampled to 100 Hz, as were the deconvolved $\Delta F/F$. The following procedures were used to extract the licking and jaw movements from the predicted positions. DeepLabCut outputs the predicted position and its confidence as a likelihood value of 0–1. This value is lower if the target body part is not visible (e.g., the tip of the tongue when the mouth is closed). However, in the model we trained, the tip of the tongue showed a high likelihood (>0.95) even when the mouth was slightly open and the tongue did not protrude from it. To separate these small mouth movements from the large tongue movements during licking of the spout, we defined the licking time point as that when the predicted X (horizontal) position of the tip of the tongue extended above 80 percent of the distance between the predicted X position of the lower lip and the predicted X position of the tip of the spout (Figure S5A). Jaw movement was defined as each time point at which a local maximum (peak) was taken from the predicted Y (vertical) position of the jaw (MATLAB's *findpeaks* function; distance between peaks, 50 ms; minimum peak prominence, 5 pixels; Figure S5B). To plot the time courses of the licking and jaw movement (Figure S5C), the frequency of the licking and jaw movement during a period from 125 ms before to 125 ms after each time point was calculated. For the nose and forelimb movements, the square root of the sum of squares of the predicted positional displacements was used. The grip position of the right forelimb was defined as the predicted lateral position of the forelimb.

Classification and quantification of movements

A successful lever-pull onset was defined as the moment when the lever trajectory exceeded a threshold (20% of the maximum displacement) in the successful lever-pull trials. In the ET trials, the time between the cue-onset and the successful lever-pull onset was defined as the reaction time. Trials with reaction times that were very short (<50 ms; 2.5% of all successful ET trials; 2.1 ± 0.4 trials, $n = 129$ sessions) were excluded from the analysis because these responses were not thought to be evoked by the sound cue. The waiting time in the IG trial was defined as the time between the successful lever-pull onset and the last time the lever trajectory dropped below the threshold. To estimate the fine-skilled lever-pull movement, we quantified the pull speed, pull duration, and pull reproducibility. The pull speed was defined as the maximum velocity of the lever trajectory between -0.2 and 0.2 s from the

successful lever-pull onset. The total time above the threshold during 3 s after the successful lever-pull onset was defined as the pull duration. The pull reproducibility was defined as the average value of the correlation of the lever trajectory of each trial to the trial-averaged lever trajectory from -0.2 to 2 s of the successful lever-pull onset (Tanaka et al., 2018; Terada et al., 2018).

Classification of neuron types

First, we defined “task-related” neurons as the neurons in which the mean values of the deconvolved $\Delta F/F$ during the movement-related period (-0.4 to 2 s from the successful lever-pull onset; F_{mov}) were significantly different from those in the baseline period (-2 to -1 s from the successful lever-pull onset; F_{base} ; $p < 0.01$; Wilcoxon rank sum test). The other neurons were classified into “task-unrelated”. The task-related neurons were further classified into increase or decrease neurons according to whether F_{mov} was higher or lower than F_{base} , respectively. For each of the increase and decrease neurons, if F_{mov} was not significantly different between the IG and ET trials ($p > 0.01$; Wilcoxon rank sum test), it was classified as a “common neuron”. If the increase or decrease neuron had higher F_{mov} in IG trials than in ET trials, it was classified as an “IG neuron” or “ET neuron”, respectively. If the increase or decrease neuron had higher F_{mov} in ET trials than in IG trials, it was classified as an “ET neuron” or “IG neuron”, respectively. The peak timing was defined as the time when the maximum (for the increase neuron) or minimum (for the decrease neuron) value was obtained in the trial-averaged deconvolved $\Delta F/F$ in the period from 2 s before to 3 s after the successful lever-pull onset. In the case of common neurons, the average over both IG and ET trials was used, and in the case of IG and ET neurons, only IG and ET trials were used, respectively. The time variation of trial-type selectivity (Figures 2D, 2E and S4K–S4N) was defined as $\text{Selectivity} = F_{\text{IG}} - F_{\text{ET}}$ for the increase neurons, and $F_{\text{ET}} - F_{\text{IG}}$ for the decrease neurons, where F_{IG} and F_{ET} are the trial average of z-scored deconvolved $\Delta F/F$ during the period from 2 s before to 3 s after the successful lever-pull onset in IG and ET trials, respectively. To plot the averaged selectivity for each trial type (Figure 2F), we calculated the average selectivity of neurons that preferred each trial type. The values of ET trials were inverted for this plot. To test for spatial clustering of IG and ET neurons, the pairwise distance was calculated as the Euclidean distance for each combination of IG and IG, ET and ET, and IG and ET neurons.

Trial-type classification analysis

Trial-type classification analysis was performed with a linear support vector machine (SVM) classifier (MATLAB’s *fitsvm* function) with 10-fold cross-validation. The classifier was trained to solve a two-class classification problem to predict trial types (IG or ET trials) from neuronal population activity. Trials with successful lever-pulls were used in the analysis, and trial types with a large number of trials were randomly subsampled so that the number of trials would be the same as those trial types with a small number of trials. The z-scored deconvolved $\Delta F/F$ of 60 randomly sampled neurons averaged in the pull-onset window (from 0.4 s before to 0.4 s after the pull onset) or 200 ms moving windows (from 2 s before to 3 s after the pull onset, every 100 ms) were used as the neuronal population activity. Ten-fold cross-validation was performed by splitting the data into ten segments using stratified partitioning. The classifier was then trained on the combined data of nine segments and tested on the remaining segment, with the process being repeated until all segments were tested. To obtain a good model, the kernel scale γ , an SVM hyperparameter, was determined from a range between 10^{-3} and 10^3 by cross-validation with the training data. Another hyperparameter, the box constraint C , was set to 1 (Tsao et al., 2018). The predicted trial types for all segments were combined and called the predicted trial types. The percentage of agreement between the predicted trial types and the actual trial types was then used as the trial-type classification accuracy (TCA). To obtain a single TCA from random subsampling, the subsampling of trials and neurons was repeated 100 times and the mean of the TCAs was used. To calculate TCA from body movements, five body movements (licking, movements of jaw, nose, and right forelimb, and grip position of right forelimb) and lever trajectory and speed (absolute value of the difference in lever trajectory) were used instead of neuronal population activity. In the experiment in which the same L2/3 neurons were tracked (Figure S9), the classifier was built using all the pursued neurons without subsampling. Decoding within the same session was evaluated by 10-fold cross-validation. For across-session decoding, decoders were created in each session, and the created decoders were evaluated by comparing their predictions with the trials in each of the other sessions.

The cue presentation in ET trials was included in the pull-onset window (the reaction time was ~ 100 – 200 ms). Thus, the success rate and reaction time in ET trials were assumed to be related to the neuronal activity during the pull-onset window. By contrast, the activity related to the waiting time in IG trials started before the pull-onset window (Figures S3A and S3B). The waiting time was related to the number of successful IG trials; thus we considered that it was not necessary for the success number in IG trials to be related to the TCA that started to increase ~ 0.5 s before the lever-pull onset. Therefore, when the TCAs calculated from the activity during the pull-onset window (TCA_{M1} and $\text{TCA}_{\text{M2} \rightarrow \text{M1}}$) were compared with the task performance, we only focused on the success rate and reaction time in ET trials. TCA monotonically decreased from the lever-pull onset to the lever-pull end; thus TCA_{M1} and $\text{TCA}_{\text{M2} \rightarrow \text{M1}}$ were assumed to also represent the context-dependent activity during the lever-pull movement. Therefore, when TCA_{M1} and $\text{TCA}_{\text{M2} \rightarrow \text{M1}}$ were compared to pull speed, pull duration, and pull reproducibility, both IG and ET trials were used (Figure 4).

Control analysis of the orofacial movements

To assess the impact of orofacial movements on the TCA calculation, we defined the neurons that represented any of the three orofacial movements: nose movement, licking, and jaw movement, as pre-pull orofacial movement-related neurons. Because trial-type differences in orofacial movements were most pronounced in the pre-pull period, we analyzed the neural activity from -1.0 to -0.2 s of the lever-pull onset. In each neuron, the mean activity and mean value of each movement during this period were calculated for all

successful lever-pulls in ET trials. The correlation between the two was calculated using Spearman's rank correlation, and neurons for which the square of the correlation coefficient obtained was greater than 0.1 were classified as neurons that represented the corresponding pre-pull orofacial movement. The contributions of pre-pull orofacial movement-related neurons, IG neurons, and ET neurons to $TCA_{M2 \rightarrow M1}$ and TCA_{M1} were analyzed as follows. First, the trial-type classifier was trained using training data, as described above. Then, the activity of each type (or multiple types) of neuron in the test data was shuffled across trials before trial-type predictions were performed. TCAs were calculated from the labels predicted from the shuffled data and compared with TCAs without shuffling.

Quantification of geometric changes in neuronal population activity

To characterize the geometric changes in neuronal population activity underlying the TCA change, we quantified the distance of the centroids of the neuronal population activity between IG and ET trials (D_{IG-ET}) and the variance of the neuronal population activity within each type of IG and ET trial (V_{IG} and V_{ET}). The z-scored deconvolved $\Delta F/F$ values averaged in the pull-onset window were used as the neuronal population activity. Then, we defined

$$D_{IG-ET} = d(\bar{\mathbf{X}}_{IG}, \bar{\mathbf{X}}_{ET})$$

where $\bar{\mathbf{X}}_{IG}$ and $\bar{\mathbf{X}}_{ET}$ are the trial average of the neuronal population activity vector in IG and ET trials, respectively, and $d(A, B)$ is the cosine distance between the vectors A and B defined as

$$d(A, B) = 1 - \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

where A_i and B_i are components of vectors A and B , respectively (Hirokawa et al., 2019; Rossi-Pool et al., 2017). This measure, like the correlation coefficient, focuses on the angle between vectors, but always takes a positive value. This ranges from 0 to 2, with 0 being the most similar and 2 being the most dissimilar. To estimate the trial-to-trial variability of neuronal population activity, we defined the variance of the neuronal population activity as

$$V_{IG} = \frac{1}{N_{IG}} \sum_{i \in IG} d(\mathbf{X}_i, \bar{\mathbf{X}}_{IG}), V_{ET} = \frac{1}{N_{ET}} \sum_{i \in ET} d(\mathbf{X}_i, \bar{\mathbf{X}}_{ET})$$

where $\mathbf{X}_i$ is the neuronal population activity vector of the i^{th} trial, and N_{IG} and N_{ET} are the number of IG and ET trials, respectively.

Encoding model

The deconvolved $\Delta F/F$ of each neuron was modeled using a linear model with inputs of motion information and neuronal population activity in different regions (Engelhard et al., 2019; Kondo and Matsuzaki, 2021; Musall et al., 2019; Park et al., 2014; Steinmetz et al., 2019). The first step was to eventize the motions. We used lever trajectory, licking, and jaw movement as the predictors, and downsampled them to 10 Hz to match the neuronal activity recordings. As licking and jaw movement were already eventized, these variables were downsampled by summing within each bin. For the lever trajectory, after calculating the mean within each bin, the time when the trajectory exceeded the threshold was defined as lever-pull, the time when the trajectory returned below the threshold was defined as lever return, and the time in between was defined as lever hold (Figure S8A). To ensure that the model fitted the trial-to-trial variability of lever speed, we added the pull speed and return speed to the variables. The pull speed and return speed were defined as the maximum or minimum value of the temporal difference of the lever trajectory (before downsampling) between -0.2 and 0.2 s from lever-pull or lever return events, respectively. The eventized motion predictors were extended backward and forward in time using an event kernel to represent the neuronal activities corresponding to motor preparation, motor execution, and sensory feedback (Hira et al., 2013; Masamizu et al., 2014). This was performed by making copies of each eventized motion predictor and shifting them temporally from 5 time points before to 10 time points after (-0.5 s–1 s) the original time point, and also including these copies as predictors. The neuronal activity we observed in this study is characterized by the fact that the same movement can exhibit different responses depending on the context. To model this feature, the number of motion predictors was further doubled so that each block could be weighted separately, with one predictor taking a non-zero value only in the IG block and another in the ET block (Figure S8A). We used the period from 2 s before to 3 s after the successful lever-pull onset for our analysis. We used ridge regression to fit the model, to avoid overfitting (Musall et al., 2019). Given the distribution of the deconvolved $\Delta F/F$, a generalized linear model (GLM) with a Poisson distribution was expected to provide the best fit (Park et al., 2014). However, when we tested this on a randomly chosen subset of the data ($n = 5$ sessions), there was little improvement in fit with a regularized GLM (elastic net) compared with the ridge regression (0.1018 ± 0.0028 and 0.1031 ± 0.0026 , respectively; $n = 1794$ neurons from M2 L2/3 and M1 L2/3). Because of the high computational cost of elastic nets, including the optimization of hyperparameters, we used ridge regression for our analysis.

For the evaluation of fitting, we used the coefficients of determination (R^2) determined by 10-fold cross-validation. R^2 was defined according to the following formula

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

where at each time point i , y_i is the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the average of the actual values over the entire time. To avoid the possibility of information leakage due to autocorrelation, the data were split into small intervals of one second in length, and from these small intervals the data were divided into training data and test data. The predicted deconvolved $\Delta F/F$ was obtained by concatenating the predicted results for all test data, and R^2 was calculated between the predicted and actual deconvolved $\Delta F/F$. Neurons with R^2 of less than 0.02 were excluded from the analysis (Steinmetz et al., 2019). The mean proportions of the excluded neurons in M2 L2/3 neurons, M2 $\rightarrow$ M1 axons, and M1 L2/3 neurons were 0.13 [95% CI: 0.11–0.15], 0.18 [0.13–0.22], and 0.086 [0.060–0.13] ($n = 40$, 17, and 40 sessions from six, four, and six mice, respectively), respectively, for IG neurons; 0.11 [0.09–0.12], 0.20 [0.14–0.26], and 0.12 [0.08–0.18] for ET neurons; 0.33 [0.30–0.38], 0.40 [0.36–0.47], and 0.30 [0.25–0.36] for common neurons; and 0.63 [0.59–0.68], 0.71 [0.64–0.77], and 0.69 [0.62–0.76] for task-unrelated neurons (Figure S8C). To analyze the contribution of each variable to neuronal activity, we used the unique contribution value (Musall et al., 2019). For the full model, we used fourteen variables (IG and ET types of lever-pull, lever return, lever hold, pull speed, return speed, licking, and jaw movement). For the reduced model to allow calculation of the unique contribution of each motion-related variable, we excluded both types of the five lever-related variables, both types of licking, or both types of jaw movement. For the reduced model used to calculate the unique contribution of the trial type, we used undoubted lever-pull, lever return, lever hold, pull speed, return speed, licking, and jaw movement. We then trained each model and calculated the difference in R^2 between the full model and each reduced model.

To investigate the influence of M2 L2/3 population activity on M1 L2/3, the activity of individual M2 L2/3 neurons was introduced into the model as a predictor. We resampled the deconvolved $\Delta F/F$ to 10 Hz and z-scored the values to avoid the effect of possible outliers such as neurons with an exceptionally high activity rate (Cunningham and Yu, 2014; Terada et al., 2018). We performed dimensionality reduction by applying principal component (PC) analysis (Perich et al., 2018; Semedo et al., 2019) to the M2 L2/3 neurons. The number of dimensions was determined for each model by searching for the number of dimensions between 1 and 35 that maximized the cross-validated R^2 of the prediction model including the population activity ($N_{\text{dim}} = 14.5 \pm 0.9$; $n = 40$ sessions from six mice). This value is comparable to the dimensionalities previously reported in the monkey premotor cortex ($N_{\text{dim}} = 14$) (Perich et al., 2018). The dimensionality-reduced neuronal population activity was added to the predictors by making time-shifted copies up to 10 time points later to allow for functional interaction up to 1 s in the past (Runyan et al., 2017). To estimate the strength of the interaction between regions, we defined the coupling index (Runyan et al., 2017) as follows:

$$\text{Coupling index} = \frac{R^2_{\text{Behaviour} + \text{M2}} - R^2_{\text{Behaviour}}}{R^2_{\text{Behaviour} + \text{M2}}}$$

where $R^2_{\text{Behaviour} + \text{M2}}$ and $R^2_{\text{Behaviour}}$ are R^2 for the full model and the model without M2 population activity, respectively. For the analysis of M2 neuron type-dependent interactions (Figure 7D), we compared a model using IG and ET neurons as sources of the neuronal population activity (“IGET model”) with a model using common neurons (“common model”). The number of PC dimensions used in each model were determined again, and were found to be 9.3 ± 0.8 for the IGET model and 9.0 ± 0.7 for the common model ($n = 40$ sessions from six mice), values that were lower than when all neuron types were used. Predictions were made with each model, and the difference in the coupling index ($\Delta_{\text{IGET-common}}$) was plotted against TCA_{M1} .

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed using MATLAB’s statistics and machine learning toolbox. Data are presented as the mean $\pm$ standard error of the mean, and error bars in graphs represent the standard error of the mean unless otherwise stated. Wilcoxon rank sum tests, paired t-tests, unpaired t-tests, F -tests, and Kolmogorov-Smirnov tests were used for statistical comparisons. Pearson’s correlation coefficient was used to calculate the correlation between two conditions. To obtain 95% confidence intervals for correlation coefficients or proportion of neurons, bootstrap resampling was performed 5000 times and 95% bootstrap confidence intervals were calculated. All of statistical details of experiments can be found in the Results section and figure legends. p -values for figures are described in the corresponding figures or figure legends. Statistical methods were not employed to predetermine sample sizes; however, sample sizes were estimated according to previous methodologically-comparable laboratory experiments and are similar to those generally employed in the field. No blinding or randomization was performed in this study.